

Review

An updated review on multi-carrier energy systems with electricity, gas, and water energy sources



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ABSTRACT

Recently, interdependency between electricity, gas, and water infrastructures are increasing due to convincing reasons and critical dependence of societies to such infrastructures. In this paper, the interconnection of the natural gas, water and power generation systems is firstly described. Then, mathematical problem formulation of economic environmental dispatch as well as deterministic/probabilistic optimization of gas/water/power nexus models are presented with emphasis on their advantages/disadvantages and achievements. Moreover, different types of seawater desalination technologies and hydro/hydrothermal/pumped hydro storages are categorized and compared in short-term scheduling and planning studies. The expected outcomes and influence of the research will be attaining a comprehensive analysis of interactions among energy sources in multi-carrier energy systems as well as enhancement of performance of the integrated power, gas and water systems in terms of operation, economy and emission of pollutant gases including significant projects implemented around the world and case studies. The potential to increase efficiency, flexibility, and performance by considering multiple energy carriers are investigated in this study to highlight the remarkable opportunities that can be attained by investing in multi-energy generation assets. The current literature is categorized into three classifications including a) integrated electricity and gas sources, b) integrated electricity and water sources, and c) integrated electricity, gas and water infrastructures. Each of the mentioned multi-carrier energy systems is introduced by concentrating on the main contributions of the literature.

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Contents

1. Introduction	3
1.1. Electricity and gas	3
1.2. Electricity and water	3
1.3. Natural gas, electricity, and water	4
1.4. Paper contributions	4
1.5. Paper organization	5
2. Economic emission dispatch of multi-carrier energy systems with natural gas and electricity energy sources	5
2.1. Problem formulation	5

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2.2.	Coupled optimization of electricity and natural gas system	7
2.2.1.	Deterministic gas-power nexus optimization methods	7
2.2.2.	Probabilistic co-optimization methods	8
3.	Multi-carrier energy systems with electricity and water energy sources	8
3.1.	Water desalination systems	8
3.2.	Hydropower plants	12
3.2.1.	Power generation of hydro plants	12
3.2.2.	Short-term hydrothermal scheduling	13
3.3.	Scheduling and planning of hydro-based power plants	13
3.3.1.	Scheduling and planning of hydro units	13
3.3.2.	Scheduling and planning of hydrothermal units	14
3.3.3.	Scheduling and planning of pumped storage units	14
4.	Multi-carrier energy systems with integration of electricity, water and gas sources	14
5.	Conclusion and future trends	16
	Declaration of competing interest	16
	Acknowledgments	16
	References	16

Nomenclature		
λ_{gas}^t	Natural gas price (\$/MWh)	E_0 Gross generated electrical energy (MWh)
N_{GL}	Number of gas-load buses	P_0 Gross generated hydropower (MW)
$G_{d,t}$	Gas demand (m^3)	H Pressure head (m)
N_{Comp}	Number of gas compressors	Q Volume flow rate (m^3)
$G_{c,t}$	Gas consumption of compressor c (m^3)	ρ Water density (kg/m^3)
N_{Gen}	Number of gas and coal fired units ($N_{Gen} = N_{Gas} + N_{Coal}$)	g Gravity (m/s^2)
a_m, b_m, c_m	Constant indices in fuel-cost characteristic of plant m (\$, \$/MWh, and \$/(MWh) ² , respectively)	η_h Hydraulic efficiency
$P_{m,t}$	Output electrical power of generating unit m (MW)	η_m Mechanical efficiency
$e_{m,fm}$	Fuel cost coefficient of power plant m due to valve point effect	η Total efficiency of hydropower unit
p_m^{min}	Minimum power production of unit m (MW)	P_R, P_W Runner/water power (MW)
$\alpha_m, \beta_m, \gamma_m, \mu_m, \delta_m$	Emission characteristic coefficients of power plant m	P_S Available power at shaft (MW)
$P_{g,t}, P_{c,t}$	Electrical power output of gas and coal fired units, respectively (MW)	E, P Available electrical energy/related hydropower (MWh, MW)
P_d^t	Electricity demand (MW)	N_p The number of thermal plants
P_{Loss}^t	Transmission losses (MW)	N_h The number of hydro units
p_m^{min}, p_m^{max}	Minimum and maximum power generation of plant m (MW)	P_{Load}^t The power load demand at each time interval (MW)
UR_m, DR_m	Ramp up and down rates of plant m (MW/h)	P_k^t Power supply of the k th hydro plant at time t (MW)
$G_{Sg}^{min}, G_{Sg}^{max}, G_{Sg}^S$	Minimum and maximum gas supply and output of gas well g , respectively (m^3)	Q_k^t Water release of the k th hydro reservoir at time t (m^3)
N_{pl}	Number of gas pipelines	V_k^t Water volume of the k th hydro reservoir at time t (m^3)
$G_{l,t}$	Gas flow rate of pipeline l (m^3)	$C_{1k}, C_{2k}, C_{3k}, C_{4k}, C_{5k}, C_{6k}$ The power generation coefficients of k th hydro unit
$\pi_{i,t}, \pi_{j,t}$	Gas pressure of nodes i and j (bar)	p_k^{min}/p_k^{max} Minimum/maximum power production of the k th hydro unit (MW)
C_{ij}	Constant factor of pipeline l between nodes i and j	V_k^{min}/V_k^{max} Minimum and maximum reservoir volume of the k th hydro unit (m^3)
T_0	Standard temperature (520°R)	Q_k^{min}/Q_k^{max} Lower/upper bounds of water release of the k th hydro plant (m^3)
π_0	Standard pressure (1 bar)	I_k^t Inflow rate of the k th hydro plant (m^3)
D_{ij}	Internal diameter of pipeline i to j (m)	S_k^t Spillage of the k th hydro plant (m^3)
L_{ij}	Length of pipeline i to j (m)	V_k^{init}/V_k^{end} Initial/final water volume of the k th hydro plant (m^3)
F_{ij}	Friction constant of pipeline i to j	V_k^0/V_k^{24} Water volume of the k th hydro plant at time intervals 0/24 (m^3)
$H_{c,t}$	Horsepower of gas compressor c (hp)	$C_i^t(P_i^t)$ Operation cost of the i th thermal unit at time t (\$)
$G_{c,t}$	Natural gas flow consumed by compressor c (m^3)	P_i^t Power supplied by i th thermal unit at time t (MW)
g_1^c, g_2^c, g_3^c	Gas consumption coefficients for compression station c (in m^3 , m^3/hp , and m^3/hp^3 , respectively)	a_i, b_i, c_i, e_i, f_i The cost coefficients of i th thermal unit
$G_{f,t}$	Gas consumption of gas-fired power producer f (m^3)	p_i^{min}/p_i^{max} Lower/upper power production bounds of the i th thermal (MW)
k_{2f}, k_{1f}, k_{0f}	Gas utilization coefficients of gas-fired power producer f (in MW/ m^6 , MW/ m^3 , and MW, respectively)	
		Indices
		t Time (hour)
		m Thermal power production units
		g Gas fired power producers

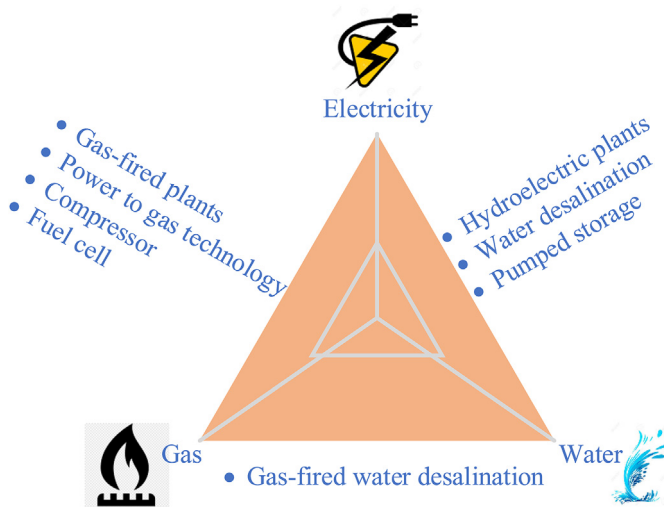


Fig. 1. An overview on multi-carrier energy systems.

1. Introduction

Nowadays, energy operation and planning are encountered with a lot of challenging issues due to the continuous increase in energy demand, finite sources of fossil fuels and global trends to use sustainable energy sources (Sebri, 2015). In addition, the reduction of energy resources, significant changes in climate status and emissions of pollutant gases are defined as basic elements for sustainable development (Rozali et al., 2014). In such an area, multi-carrier energy systems, which are defined as a combination of two or several energy conversion tools, are introduced as a practical solution for the above-mentioned challenges (Fathima and Palanisamy, 2015). The integration of electricity, gas, and water energy sources has attracted researchers' attention in recent years considering a series of global, economic and environmental advantages. Several studies have been concentrated on the interdependency between a) electricity and gas, b) electricity and water and c) electricity, gas, and water-based energy systems, which is shown in Fig. 1.

1.1. Electricity and gas

The electrical energy is generated by a combination of different primary energy sources including coal, natural gas, nuclear energy, and renewables. The variability and uncertainty characteristics of electricity generation of renewable energy sources require a more flexible power generation system. Interdependency between electricity and gas appears by gas-fired generation, which are effective in demand of both systems, are practical solutions to overcome the above-mentioned challenges (Mesfun et al., 2017). The fluctuations of electricity supply in power systems can be exchanged to gas networks and gas infrastructure appearing in a series of issues (Venetsanos et al., 2002). Such interdependency between gas and electricity networks is seen as gas load in gas systems and as electricity providing sources in energy networks. The electricity generation by gas-fired plants, which transform chemical energy into electric form takes advantage of quick responses, less emission and cost savings. The annual energy outlook 2014 from US energy information administration reported the allocation of gas load in electricity generation as 232.22 Gm³, which is aimed to increase up to 266.21 Gm³ by 2040. Additionally, it is reported that the increased rate of gas demand will be 2.9–3.2% per year until 2030 (Üster and Dilaveroglu, 2014). The combined heat and power (CHP) units (Nazari-Heris and Mohammadi-Ivatloo, 2017), combined cooling heating and power (CCHP) units (Jabari et al., 2016) and distributed heating and cooling loads can be introduced as other examples of coupling gas and electricity networks. In a general overview, the interactions between such two networks play a significant role in the area of global energy demand-supply. According to annual energy outlook 2020, the oil and natural gas plant liquids production in U.S. will change by 2050 as shown in Fig. 2 (Center, 2020). It is obvious that production of oil and natural gas plant liquids continues to grow through 2025. After 2025, it may remain almost constant and even be reduced. Therefore, optimal planning of integrated natural gas and power networks is necessary to manage these limited energy resources.

1.2. Electricity and water

Electricity and water can be introduced as essential entities for human life. The increase in the population of the world, improvements in agriculture and industries and major revisions in

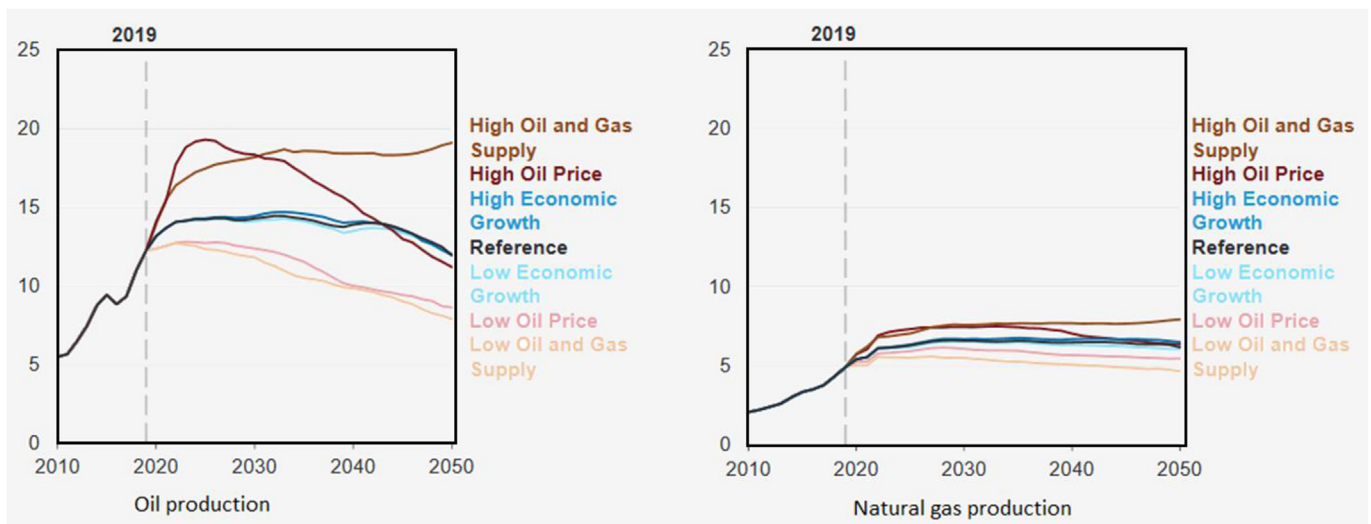


Fig. 2. U.S. oil and natural gas production patterns in million barrels per day until 2050 (Center, 2020).

standards of human life have the most important effect on electricity and water scarcities (Cominola et al., 2018). Considering the reported global statistics, 97.5% saltwater has been covered in the earth, which is a significant source for providing fresh water for the world population (Shannon et al., 2008). One of the practical solutions to such trouble is defined as water desalination systems, which is an intensive energy-consuming process. In such an area, the combination of renewable energy sources with water desalination systems has become attractive, for which several remarkable efforts have been carried out. Unlike renewable desalination systems, the conventional thermal desalination systems resulted to emission of harmful gases. The renewable desalination system has considerable advantages of economic viewpoints, social development and ecological environment. Such a concept can be introduced as a green and environmentally sustainable process (Chandrashekar and Yadav, 2017). The above-mentioned importance of energy source for water desalination systems highlights a deep analysis of in the integration and design of renewable energy-based desalination technologies. Also, very limited studies have reported on the environmental impact of water desalination systems in the worlds, which shows the importance of a study and investigations of the associated emission factors. In the area of integration of electricity and water, hydro-based generation units such as hydrothermal systems are incorporated in providing the load demand of the system (Nazari-Heris et al., 2017b). Such generation concepts are effective in satisfying the electricity demand considering the ignorable operational cost of hydro units (Nazari-Heris et al., 2017a). On the other hand, considerable water is consumed for the generation of electricity by hydro-based plants. For instance, the National Renewable Energy Laboratory (NREL) has reported 68.9 L/kWh for water consumption in hydropower in the United States (Torcellini et al., 2003). It is worth to mention that pumped hydro storage systems are 96% of energy storage technologies in power systems. Such hydro storage technologies can respond to power shortage of grids in seconds/minutes, which is effective in enhancing the resiliency of systems, which highlights the need for analysis in this area. Investments in pumped storage systems can greatly improve the resiliency of integrated system and demand supplies showing the importance of analysis of the level of combination of such units in energy systems.

1.3. Natural gas, electricity, and water

As we know, the conventional gas-fired power plants participate in supplying the total electricity requirement of seawater desalination systems. Meanwhile, world potable water demand is forecasted to increase by up to 60 Gm³ per year by 2050. Moreover, all desalination plants consume 75 TWh per year resulting in yearly 220 Mt CO₂ emissions by 2040 (Shahzad et al., 2017). These water-energy-environmental problems can be overcome using renewable and sustainable energy sources such as solar, wind, hydro, geothermal, and etc. to produce electricity for seawater desalination processes. Hence, the optimal operation of multi-carrier energy systems with gas, electricity and water resources will be more and more urgent to deal with associated water and energy crisis (Quteishat, 2009). Currently, Saudi Arabia with 27 desalination plants produces the largest volume of freshwater (3.3 Mm³/day) using two multi-stage flash distillation (Al Ghamdi and Mustafa, 2016) and multi-effect distillation (Al-Mutaz and Wazeer, 2014) technologies which both are integrated with generation stations. It is predicted that India's population will increase up to 1.60 G by 2050 and ranks it 40th in the world from electricity and freshwater scarcity viewpoint (Manju and Sagar, 2017). Therefore, the highest daily global solar irradiation in India (4–7 kWh/m²) can be utilized for electricity generation and solar desalination processes (Sahoo

et al., 2015). In China, a successful experience of Caofeidian industrial zone on optimal operating of multi-carrier energy systems with gas, electricity, and water energy sources, demonstrated that by annually 18 Mm³ of seawater desalination, 600,000 t salts were harvested, 10 Mm³ potable water was saved, and 40,000 t of CO₂ emissions was mitigated (Zou and Liu, 2016). Regardless of considerable efforts on studying integrated energy systems, limited number of works have focused on developing a comprehensive model for integrated power, water, and gas systems. It is important to develop a model to determine modeling of the operation of such integrated systems and the function of interconnecting elements among different systems. Also, the role of integration of water, gas and power systems in satisfying a desirable performance of the whole system in terms of operation cost, emission of pollutant gases, and resiliency of the whole integrated energy system should be investigated.

1.4. Paper contributions

As mentioned, there are several interesting works on modeling, operation, scheduling, planning, and optimization of integrated water-natural gas-power generation systems. But, a comprehensive literature review about the presented approaches and the advantages/disadvantages of the proposed methods has not been provided by scholars. Moreover, the interconnection of cogeneration grids (for example gas-power nexus or water-power hybrid networks) and trigeneration ones as well as the different probabilistic approaches used for modeling the uncertainties associated with gas loads, potable water demand, electricity consumption, energy market prices, renewable energy resources, etc. have not been compared yet. Considering significant research in the area of multi-carrier energy networks with gas, water and power sources, a comprehensive review for such networks remains untapped.

This paper aims to provide a comprehensive updated review on a combination of electricity, gas, and water energy sources as multi-carrier energy systems. The specific contribution of this project will be providing a comprehensive review on operation model and performance assessment of interconnectivity in multi-carrier energy networks with gas, power, and water sources as well as interconnecting elements among different energy carriers such as emerging energy storage technologies and renewable energy sources. The main objective of this research is to provide a comprehensive investigation of interactions among gas, power, and water energy sources in multi-carrier energy networks, and evaluating the performance of the integrated power, gas and water systems from economy and emission viewpoints. The analysis will be performed based on real case studies in the world and test systems introduced for multi-carrier energy systems. There are several challenges in the area of consideration of interconnection of power, gas, and water energy carriers including the effect of considering multi-carrier energy systems on the cost of supplying multi-carrier energy demands and emission of pollutant gases, evaluating the capability of storing one energy type in another one such as power to gas technology and hence improving the performance of the integrated system, and the modeling of interconnectivity among power, gas, and water energy carriers in an energy system. Accordingly, the introduction to each system will be provided by concentrating on the most contributions and considerations of studies reported in such areas. In addition, the implementation of different electricity, gas and water-based multi-carrier energy systems will be compared with respect to each other from different points of view. In addition, different co-optimization strategies for solving the economic environmental dispatch problem of multi-carrier energy systems with natural gas, electricity, and water energy sources are provided. Hence, objective function

and operational limits in coupled optimization of the natural gas network, electricity grid, hydropower plants, and sea water desalination processes are presented. Additionally, brief definitions and comparisons between the most popular optimization methods for economic environmental dispatching of multi-carrier energy systems with gas, water, and electricity energy hubs are provided. The important projects accomplished around the world on the integration of gas, water, and electricity systems are proposed and investigated in this study. The expected outcomes and influence of the research will be attaining a comprehensive review on interactions among energy sources in multi-carrier energy systems as well as enhancement of performance of the integrated power, gas and water systems in terms of resiliency, operation cost and emission of pollutant gases.

1.5. Paper organization

The remainder of this paper is set out as follows: Sections 2 represents a comprehensive review of optimal operating and scheduling of natural gas and electricity comparing the introduced coupled optimization approaches. In addition, the current literature on multi-carrier energy systems with electricity and water energy sources is investigated in Section 3. Then, Section 4 has focused on multi-carrier energy systems with natural gas-electricity-water

comparison with traditional coal-fired processes (Martinez-Mares and Fuerte-Esquivel, 2012). The natural gas and power nexus are optimized to minimize the fuel cost and emissions of the producers considering the operational limits of gas and power networks (Qadrdan et al., 2017). As illustrated in Fig. 3, a typical integrated natural gas and electricity grid consists of gas wells and pipelines, electricity transmission lines, gas compressors, coal and gas-fired generators, and different loads such as gas, electricity, cooling, and heating (Qiu et al., 2015a). The natural gas is generated within the gas wells and transferred via the pipelines. It then propels gas compressors and is delivered to gas loads or directly used for cooling and heating applications.

The main objectives of the economic emission dispatch problem in the combined natural gas and energy grids is to minimize the fuel cost and CO₂ emissions of the coal and gas-fired generators considering their valve point effects, which can be given by (1) and (2), respectively (Elaiw et al., 2012). In the fuel cost objective function, the first and second terms are related to the total natural gas consumed by gas compressors and gas loads, respectively. While, the fuel cost of the coal and gas-fueled power plants is expressed by a quadratic term versus output electricity (Elaiw et al., 2013). Moreover, a sinusoidal term is added to the quadratic fuel cost to model the valve point effects of the thermal generation stations.

$$\text{Fuel cost} = \min \sum_{t=1}^T \left[\underbrace{\lambda_{\text{gas}}^t \sum_{d=1}^{N_{\text{GL}}} G_{d,t}}_{\text{Gas loads cost}} + \underbrace{\lambda_{\text{gas}}^t \sum_{c=1}^{N_{\text{Comp}}} G_{c,t}}_{\text{Gas compressors cost}} + \underbrace{\sum_{m=1}^{N_{\text{Gen}}} (a_m P_{m,t}^2 + b_m P_{m,t} + c_m)}_{\text{Electricity production cost}} + \underbrace{\sum_{m=1}^{N_{\text{Gen}}} |e_m \sin(f_m (P_m^{\min} - P_{m,t}))|}_{\text{Valve point effect}} \right] \quad (1)$$

energy sources. Finally, Section 5 concludes the proposed review study on multi-carrier energy systems with different energy carriers.

2. Economic emission dispatch of multi-carrier energy systems with natural gas and electricity energy sources

2.1. Problem formulation

Recently, the interconnection of natural gas and power systems is increasing because of lower CO₂ footprints and fuel costs, and fast response of gas-fired generators to fluctuations of renewables in

$$\text{Emissions} = \min \sum_{t=1}^T \sum_{m=1}^{N_{\text{Gen}}} (\alpha_m + \beta_m P_{m,t} + \gamma_m P_{m,t}^2 + \mu_m \exp(\delta_m P_{m,t})) \quad (2)$$

Subject to:

- Electricity balance limit: As (3), the sum of energy demand and transmission losses should be equal to the power generation of coal and gas-fired units (Bai et al., 2016). where, $P_{g,t}$ and $P_{c,t}$, indicate the electricity production of the gas and coal fired

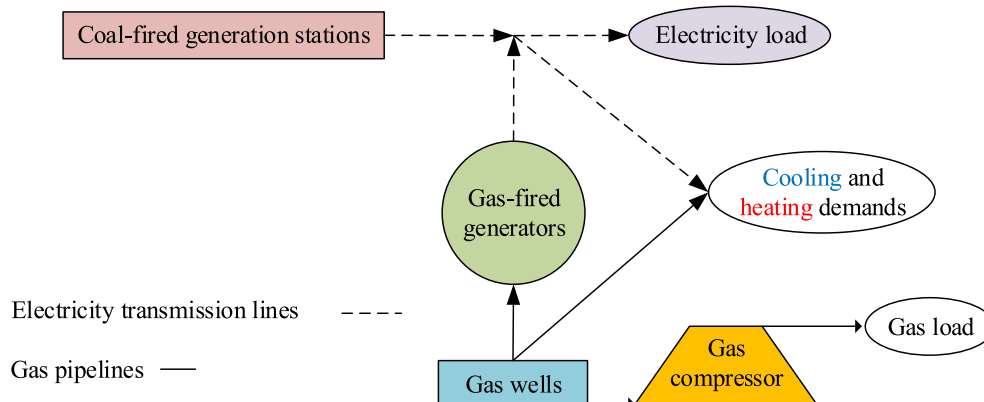


Fig. 3. The schematic presentation of the gas-power coupled systems.

plants, respectively. In addition, P_d^t and P_{Loss}^t are total electrical demand and transmission system power losses at time t , respectively. According to (4), the transmission losses is calculated by B-coefficients.

$$\sum_{g=1}^{N_{Gas}} P_{g,t} + \sum_{c=1}^{N_{Coal}} P_{c,t} = P_d^t + P_{Loss}^t \quad \forall t = 1, 2, \dots, T \quad (3)$$

$$P_{Loss}^t = \sum_{i=1}^{N_{Gen}} \sum_{j=1}^{N_{Gen}} P_{i,t} B_{ij} P_{j,t} + \sum_{i=1}^{N_{Gen}} P_{i,t} B_{0i} + B_{00} \quad \forall t = 1, 2, \dots, T \quad (4)$$

- Power generation capacity: The power production capacity of the gas and coal-fired plants is limited by minimum and maximum values, as intended in (5) (Zhang et al., 2015b).

$$P_m^{min} \leq P_{m,t} \leq P_m^{max} \quad \forall t = 1, 2, \dots, T; \quad \forall m = 1, 2, \dots, N_{Gen} \quad (5)$$

- Ramp up and ramp down of thermal generating units: The increase and decrease of the power production of unit i are limited by its ramp up and down rates, as shown in (6) (Unsihuay-Vila et al., 2010).

$$\begin{aligned} \text{Max}(P_m^{min}, P_m^{t-1} - DR_m) &\leq P_{m,t} \leq \text{Min}(P_m^{max}, P_m^{t-1} + UR_m) \\ \forall t = 1, 2, \dots, T; \quad \forall m = 1, 2, \dots, N_{Gen} \end{aligned} \quad (6)$$

- Gas wells limits: In the hybrid gas and power network, the natural gas is injected from wells or suppliers into the gas-fired thermal units, gas compressors or loads, and the heating or cooling processes. Hence, the upper and lower bounds of the gas production should be considered for each gas well as (7) (Unsihuay-Vila et al., 2010).

$$G_{S,g}^{min} \leq G_{g,t}^S \leq G_{S,g}^{max} \quad \forall t = 1, 2, \dots, T; \quad \forall g = 1, 2, \dots, N_{Gas} \quad (7)$$

- Pipeline transmission constraint: The natural gas flow rate through each pipeline depends on the physical characteristics of the pipe such as length, diameter, temperature, altitude drop, and the pressure difference between its two ends, which can be given by the following equations (Unsihuay-Vila et al., 2010).

$$\begin{aligned} G_{l,t} &= \text{sgn}(\pi_{i,t}, \pi_{j,t}) C_{ij} \sqrt{|\pi_{i,t}^2 - \pi_{j,t}^2|} \quad \forall t = 1, 2, \dots, T; \\ \forall l = 1, 2, \dots, N_{pl} \text{sgn}(\pi_{i,t}, \pi_{j,t}) &= \begin{cases} 1 & \pi_{i,t} \geq \pi_{j,t} \\ -1 & \pi_{i,t} \leq \pi_{j,t} \end{cases} \end{aligned} \quad (8)$$

where, C_{ij} depends on its physical characteristics and is calculated as relations (9) and (10) (An et al., 2003). Moreover, G is the ratio of natural gas density at standard pressure and temperature to air density at the same condition.

$$C_{ij} = 3.2387 \frac{T_0}{\pi_0} \sqrt{\frac{D_{ij}^5}{L_{ij} G F_{ij} Z_a T_{ij}}} \quad (9)$$

$$F_{ij} = \frac{0.032}{\sqrt[3]{D_{ij}}} \quad (10)$$

- Gas compressor: The gas compressor is installed between the nodes i and j to compensate for the pressure drop during the gas transmission via pipeline l . The transmitted gas flow via the pipeline l using the gas compressor c is given by (11), where, k_{c1} , k_{c2} , and α depends on structural characteristics of gas compressor c , while $H_{c,t}$ denotes the horsepower of the gas compressor c , which should be restricted by lower (H_c^{min}) and upper (H_c^{max}) limits as (12). If the compression station c is integrated with an electricity node, the local electricity grid will supply the total energy requirements of gas compressor c . Hence, $H_{c,t}$ is considered as an electrical load in the electricity power balance criterion. Otherwise, it will directly consume natural gas flow, $G_{c,t}$, to satisfy $H_{c,t}$ as expressed by (13).

$$\begin{aligned} G_{l,t} &= \text{sgn}(\pi_{i,t}, \pi_{j,t}) \frac{H_{c,t}}{k_{c2} - k_{c1} \left[\frac{\text{Max}(\pi_{i,t}, \pi_{j,t})}{\text{Min}(\pi_{i,t}, \pi_{j,t})} \right]^\alpha} \quad \forall t = 1, 2, \dots, T; \\ \forall l = 1, \dots, N_{pl} \end{aligned} \quad (11)$$

$$H_c^{min} \leq H_{c,t} \leq H_c^{max} \quad \forall t = 1, 2, \dots, T \quad (12)$$

$$G_{c,t} = g_1^c + g_2^c H_{c,t} + g_3^c H_{c,t}^2 \quad \forall t = 1, 2, \dots, T \quad (13)$$

- Gas flow balance constraints

Similar to electrical power balance criterion, a natural gas flow balance constraint is regarded at each gas node as follows:

$$G_{S,g}^S = G_{g,t} + G_{d,t} + G_{l,t} + G_{c,t} \quad \forall t = 1, 2, \dots, T \quad (14)$$

- The electricity output of gas-fired producers

Output electrical power of a gas-fired thermal unit, $P_{g,t}$, with respect to natural gas consumption is modeled using a quadratic function as (15).

$$\begin{aligned} P_{g,t} &= k_{2f} G_{g,t}^2 + k_{1f} G_{g,t} + k_{0f} \quad \forall f = 1, 2, \dots, N_{Gas}; \quad \forall t \\ &= 1, 2, \dots, T \end{aligned} \quad (15)$$

According to Fig. 4, U.S. electricity demand growth, low natural gas prices, carbon-free nature and favorable costs of renewables will make these resources more attractive through 2050. Hence, deterministic and probabilistic planning and scheduling of combined natural gas and power systems without and with incorporation of uncertainties associated with price, demand and clean energy producers are important for making the risk-aversion and risk-seeker decisions. Subsections 2.2.1 and 2.2.2 present the risk-neutral and risk-constrained approaches, which models the

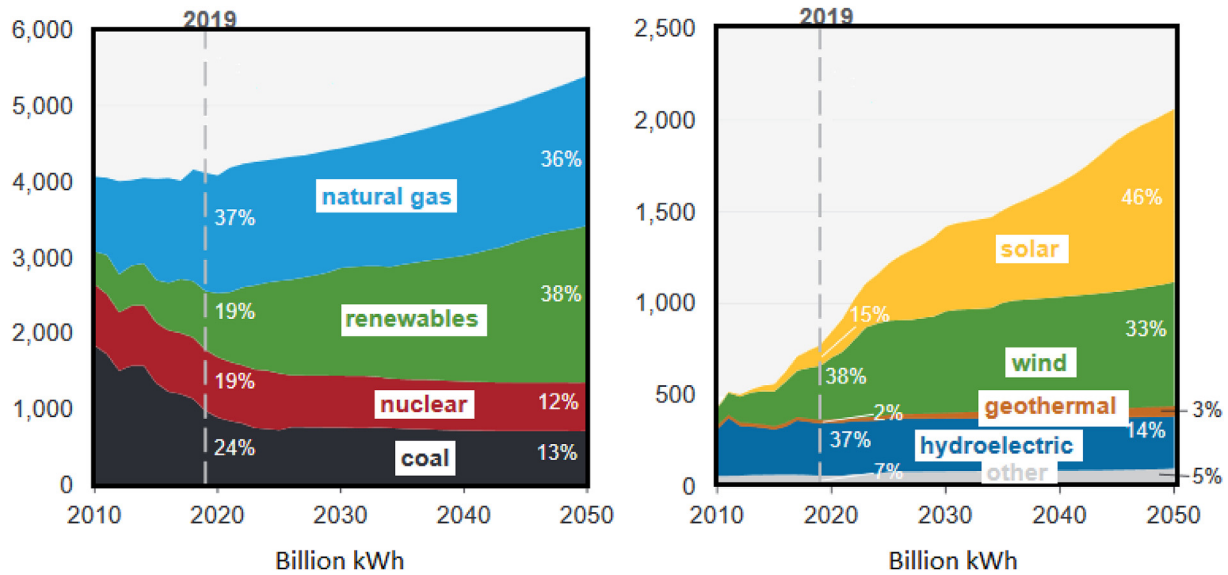


Fig. 4. Electricity generation from fuels and renewables (Center, 2020).

mentioned uncertainties in long and short-term studies.

2.2. Coupled optimization of electricity and natural gas system

2.2.1. Deterministic gas-power nexus optimization methods

The risk-neutral coordination of the integrated natural gas and electricity (NGE) system has widely been discussed using the various optimization methods. The optimal multiple gas-power hub interconnection planning algorithm is developed in Zhang et al. (2017). The minimal cut-maximal flow algorithm is used to determine an interface least-cost area composed of transmission lines and gas pipelines to find how to interconnect the energy hubs among a set of candidate paths. The total investment cost of power transmission lines and gas pipes is minimized as the objective function. Maximum energy and gas flow through interface transmission lines and pipes into each hub system should be equal to or higher than its predicted natural gas and electricity peak demands. For each energy hub grid, the average number of hours or days over a 1-year period, that the gas and power peak demands are expected to be more than the production capacity of thermal units and gas suppliers, is limited by the expected load losses. Simulation on a test system with 10 power-gas hubs demonstrated that optimal interconnection of multi-carrier hub networks is a cost-effective and more reliable approach in supplying their natural gas and electricity loads via transferring power and gas through interface lines and pipes. In this research, the economic and environmental benefits of power-to-gas technology and renewable energy resources could be considered in long-term planning (van Leeuwen and Mulder, 2018). At off-peak electrical demand hours, the surplus power obtained from the clean energy sources can be employed for driving an electrolyzer and generating hydrogen. Then, hydrogen can be combined with CO₂ to produce methane (Mehrerjedi, 2019), injected to the natural gas grid (Tsujimura and Suzuki, 2019), and employed in the transportation sector (Murugan et al., 2019).

In (Liu et al., 2010), a joint operator is responsible for optimal scheduling of natural gas and power hybrid transmission systems to minimize the operation costs of the electricity and gas networks using the augmented Lagrangian relaxation method. The operation cost of the power grid consists of the fuel cost, start-up and shut down costs of the non-gas fired power plants, penalty cost of not-

supplied power demands, the non-fuel operation cost of gas-fired thermal units. The penalty factor of the not-supplied gas loads and the operation costs of the gas wells, liquefied natural gas stage, and the gas storage unit are considered as the second term of the objective function. The Lagrangian dual is solved by a security-constrained unit commitment subproblem combined with hydro coordination and natural gas allocation subproblem. It is recommended to apply demand-side management strategies such as time-of-use demand response programs (Hosseini Imani et al., 2019), real-time forced (Jabari et al., 2019) and incentive (Eissa, 2018) based schemes, critical peak pricing (Gazijahani and Salehi, 2018), demand bidding strategy (Tarasak et al., 2014), interruptible-load and coupon-based virtual power plant models (Cui et al., 2016) on both gas and power consumers to improve the daily operation cost of the hub network. Reference (Zeng et al., 2017) focused on the coordinated expansion planning of the NGE systems considering the gas to power generators and the power to gas technology. A bi-level mixed-integer programming model is solved to maximize the investment and operation costs of the hybrid grid. The power and gas production capacities are determined in the upper-level using the modified binary particle swarm optimization (BPSO). In the lower-level, the economic dispatch of the integrated system is conducted using the outputs of the upper-level and the interior point method (IPM). The environmental issues associated with the gas-fired thermal power plants and the gas suppliers should be intended in the lower-level optimization process to solve the economic-emission dispatch problem (Liang et al., 2018). Moreover, Bender's decomposition technique should be used for developing a contingency-constrained co-expansion planning approach and model the impact of the outage distribution factors of the power transmission lines and the gas pipes in long-term analysis (Zhou et al., 2019). Erdener et al. (2014) analyzed the gas dependency of natural gas-fired generating units and electricity dependency of gas compressors as two major interdependencies between natural gas and electricity systems using a dynamic simulated model under MATLAB software. They investigated how the failures triggered in each system affect another grid. The transient and steady-state stability of the coupled power-gas systems should be assessed in DIGSILENT to improve the speed and accuracy of the proposed model. The natural gas and electricity grid co-expansion planning scheme is investigated in (Chaudry et al., 2014)

to minimize the operational and expansion costs. The gas system is expanded by increasing the diameters of the gas pipes as well as installing the new gas compressors and storages, while the electricity grid expansion is implemented by improving the transmission capacity. The fuel cost of the gas-fired thermal units could be reduced with optimal charge and discharge of advanced adiabatic compressed air energy storage at off-peak and on-peak electrical demand hours, respectively (Nakhamkin, 2012). In (Kahr et al., 2013; Katsaprakakis et al., 2015), it is studied that the lower price of natural gas, lower capital cost of gas-fired plants and less CO₂ footprints in comparison with coal-fired units makes them cost-effective for China and Greece's grids. A security-constrained bi-level economic dispatch model is proposed in (Wang et al., 2017) for short-term scheduling of joint gas and power system considering wind power and power-to-gas process. In the power to gas process, electrolyzers absorb the surplus electrical power from the wind farms to produce hydrogen, inject it into the gas pipelines in quality and quantity compatible with gas safety requirements, and reduce the natural gas volumes supplied from the gas nodes (Qadrdan et al., 2015). The reliability of the coupled system should be evaluated under the worst case study when the wind product is not sufficient and the electrolyzers cannot be driven due to the wind shortage (Salehi-Dobakhshari and Fotuhi-Firuzabad, 2011). Innovative cogeneration electricity and synthetic natural gas are designed in (Minutillo and Perna, 2014). Hydrogen is produced through the electrolysis process and then employed for synthetic natural gas production. The syngas is also used as an alternative fuel in a molten carbonate fuel cell plant for electricity generation. The chemical exergy analysis should be provided for each component. The exergy efficiency of the hydrogasification process could be compared presented. A chance-constrained mixed-integer nonlinear programming problem is solved by Odetayo et al. (2017) to find the best size and site of the gas-fired generating units and capacitor banks while minimizing the operational costs of the natural gas and electricity grids over an operating interval of ten years. The forced outage rates of the gas pipes and power transmission lines have not been modeled in the long-term optimal planning method. Moreover, corrosion repair and maintenance costs of equipment have not been considered in this optimization problem. The load growth during the 10-years period may affect the optimal power generation and gas extraction schedules of the producers. In (Sheikhi et al., 2015), it is proved that the integrated demand response program implemented on both natural gas and electricity demands decreases the energy procurement cost during on-peak load hours, significantly. From another point of view, the analysis of the gas emissions, which are produced during the electricity generation, has a great importance in investigation of combined gas and electricity systems (Khan et al., 2018).

2.2.2. Probabilistic co-optimization methods

In the literature, the coupled optimization of NGE networks has been carried out considering the uncertainty of both natural gas and electricity prices and demands, and renewable energy sources. In this context, Refs (Alabdulwahab et al., 2015; Qiao et al., 2017), studied the impacts of wind generation uncertainties on NGE systems using a stochastic security-constrained unit commitment problem (Liu et al., 2009) and an interval gas flow analysis for minimization of the expected operation cost over a 24-h time interval. An autoregressive moving average (ARMA) based Monte Carlo simulation and Benders decomposition technique are applied to generate uncertain wind speeds and evaluate the natural gas system constraints, respectively. It is demonstrated that flexibility and quick ramping capability of natural gas-fired generators compensates for the uncertainty of wind products. References (Zhang et al., 2016a, 2016b) solved the same problem considering the

random outages of generators and transmission lines, and random errors in the hourly load forecasting process. A robust co-optimization planning strategy is introduced in (He, C. et al., 2017), which finds the optimum operating points of electricity and gas networks by solving a column-and-constraint generation and outer approximation algorithm for the power system as a master problem and mixed-integer linear programming model as a sub-problem for the gas grid. This robust scheduling strategy considers operational cost and dynamics of the natural gas system, the power to gas process, and power system uncertainties. Impacts of natural gas system constraints on mid-term risk-constrained hydrothermal scheduling problem are analyzed in (Sahin et al., 2011). Moreover, uncertainties of electricity and gas market prices and natural water flows are modeled via Monte Carlo scenarios in the stochastic unit commitment problem. Deane et al. (2017) applied an integrated electricity and gas model on 28 country European power and gas systems (EU-28) to study all probable scenarios of yearly gas supply interruptions. It is proved that the interruption of Russian gas supply to the EU leads to an increase in expected gas and electricity prices of 28% and 12%, respectively. The authors in (Nazari-Heris et al., 2020) have presented a security-constrained operation model of combined gas and power systems with significant integration of wind power, for which a power to gas system is employed to decrease wind curtailment. In this reference, the uncertainties of wind power, power and gas demands are handled using stochastic programming approach. Diagoupis et al. (2016) developed a Monte Carlo simulation-based reliability assessment procedure for the robust operation of the electricity market against failure events of gas supply system. Additionally, impacts of alternative natural gas storage locations on electricity prices, unit commitment and economic dispatch problem, are studied. A multi-objective framework is presented in (Cheng et al., 2016) for optimal planning of hybrid NGE system to minimize both investment and production costs while considering N-1 system security criterion. Elitist non-dominated sorting genetic algorithm and primal-dual interior-point method combined point estimation method are used to obtain optimal Pareto solutions and evaluate impacts of wind generation uncertainties on economic emission functions, respectively. The authors have proposed a hybrid information gap decision theory-stochastic approach in (Mirzaei et al., 2020) to handle the uncertainties of power and gas loads as well as wind power for minimizing total operation cost considering the influence of power to gas system and demand response.

To present an effective review on optimal planning of integrated natural gas, electricity and water systems, other comparative researches with different objective functions and constraints are summarized in Table 1.

3. Multi-carrier energy systems with electricity and water energy sources

The multi-carrier energy systems with the integration of water and electricity energy sources are defined as an important part of human life in the form of hydropower units as well as desalination plants. This section aims to provide a combination of electricity systems and water in terms of water desalination and hydropower energy systems.

3.1. Water desalination systems

Many points of the world especially highly populated coastal areas suffer from severe drought situations. Seawater desalination is considered as the most expensive concept of supplying drinking water at the commercial scale due to the high costs associated with

Table 1

Optimization of natural gas and electricity hybrid grids using probabilistic strategies with different objective functions and constraints.

Refs.	Proposed approach and contributions	Test system	Advantage/disadvantage	Major achievements of the performed optimization
Chen et al. (2017)	Optimal energy flow for minimization of load-generation mismatch and maximization of grids' loadability	IEEE 39-node system and a 48-node gas system	Achieving an accurate optimal energy flow solvability region considering reversible gas flows	The optimization results showed that the solvability of optimal energy flow can be accurately recognized, and reversible gas flows are necessary for attaining correct solvability identification results.
Gil et al. (2016)	- Electricity side management for maximization of power system profits - Gas side management strategy for minimization of gas operation cost while determining total natural gas consumed by the electricity grid	- An electricity grid with two generation companies: Company A with 10 generators; Company B with 5 generators - A gas grid with two generation companies: Company C for satisfying conventional inelastic load and company D for gas-fired generating units	Allowing generation companies to obtain synergies in a competitive market environment	Obtaining similar solutions for electricity/gas side viewpoints is based on the precision of the information interchange, which is the equivalent contracts or marginal revenue-gas usage relationship between two cases.
(Qiu et al., 2015b; Unsuhay-Vila et al., 2010)	Sequential importance sampling-based flexible co-expansion planning method for maximization of social welfare considering some uncertainties such as demand growth, fuel cost, and financial constraints	IEEE 14-bus power system and 14-bus gas grid	- Better identification of energy infrastructure weakness for supplying total electricity demand - Social welfare improvement by co-operation of natural gas and electricity grids	The planning optimization of the integrated network expansion provides strategic consolidation recommendations for both gas and power sides.
Zhang et al. (2017)	Minimum cut-maximum flow approach to obtain a least-cost network of transmission lines and gas pipelines from a given set of candidate paths	A 10-hub system with 12 transmission lines, 30 generators, 9 gas pipelines, and 1 natural gas supplier	Reliability improvement by transmitting energy from the electricity network to the natural gas system and vice versa.	The optimization selects the least-cost of the system equipment in terms of power and gas lines resulting in a reliable link of multi-energy hubs.
Alabdulwahab et al. (2015)	- Benders decomposition technique coupled stochastic security-constrained unit commitment - Wind speed prediction errors modeling in Monte Carlo scenarios using autoregressive moving average time-series	6-bus power system with a 6-node natural gas system - IEEE 118-bus system with a 10-node natural gas system	Higher flexibility of natural gas-fired generators for satisfying not-supplied demand under lower wind productions	The optimization results verify that wind power can be effective in decreasing the dependence of power network on the constrained gas system.
Nunes et al. (2018)	Long-term integrated planning using a static stochastic cost minimization method considering uncertainties of loads and gas price	Queensland test system with 11 bus in the electricity grid and 10 nodes in the gas network	- Saving \$500 M up to 2030 using integrated stochastic analysis of wind-solar hybrid gas-power nexus model - The uncertainty associated with solar and wind products should also be considered in the stochastic study.	The optimization solution suggests that combined planning of gas and power systems results in lower the whole cost of the networks and efficiency of allocating and operation of the gas system.
Qiu et al. (2018)	- Planning of solar PV, battery, and gas-fired microturbine in power-gas nexus model using a two-stage stochastic optimization algorithm - Optimal investment in stage 1 and operation in stage 2 - Solving a mixed-integer quadratic programming problem using the k-point estimation method	14-bus coupled micro energy grid	- The application of energy storage systems in electrical grids has been discussed. - The participation of gas storage mediums such as isolated underground caverns has not been considered in the natural gas network.	The optimization model can determine investment and operation feasibly and optimally from an integrated viewpoint, which gathers the meritable solution of the planning of microgrids. - Case 1: With the single-stage optimization of the separated gas and power grids, the daily operating cost of \$94.1 is found. - Case 2: For coupled micro gas and electricity network, it is reduced to \$78.8.
(Qiu, J. et al., 2015)	Expansion co-planning approach for minimization of operation cost, carbon emission cost, and unreliability cost and modeling the uncertainties of the natural gas utilization	A 3-node gas grid and 2-bus energy network in combination with Victorian coupled gas and electricity system in Australia	Uncertainty of gas/electricity market prices, as well as fluctuations in energy demand, have not been considered in probabilistic analysis.	The optimization methodology can identify the expansion planning based on the location, time and type of gas-fired power generation units, gas pipelines and power lines.
Yuan et al. (2020)	- Stochastic analysis of an energy hub consisting of power to gas storage, combined heat and power (CHP) unit, wind power, boiler, electrical storage, and thermal storage unit - Implementation of incentive-based demand response programs in electrical sector	•Stochastic analysis of an energy hub consisting of power to gas storage, combined heat and power (CHP) unit, wind power, boiler, electrical storage, and thermal storage unit	The uncertainty of the wind products has been considered to gain from the surplus wind energy using the power to gas technology.	Reduction of operation cost by approximately 7%
Bao et al. (2019)	- The gas pipeline flow model under non-isothermal condition - Considering the hydrate formation in the gas networks	•Case 1: 13-node natural gas network composed of 3 gas sources, 3 compression stations, and 1 pressure-regulating station. This network connects to electrical grid via the nodes 6 and 8.	The pressure-regulating stations and compression stations have been considered in simulations.	The wind penetration rate affect the risks of natural gas systems significantly.

(continued on next page)

Table 1 (continued)

Refs.	Proposed approach and contributions	Test system	Advantage/disadvantage	Major achievements of the performed optimization
Hu et al. (2019)	- The probabilistic-interval energy flow has been extended in a regional level integrated electricity and gas system using the polynomial chaos expansion method. - The probabilistic/interval uncertainties, linear/rank correlations among probabilistic variables, and dependency between interval variables have been developed using the methodological methodology.	Case 2: 31-node gas grid consists of 5 gas sources, 5 compression stations, and 2 pressure-regulating stations. This system connects to energy network through the nodes 10, 24 and 31.	The PV power products should be evaluated under the clear and cloudy sky conditions.	- Much lower computation in comparison with Monte Carlo simulations (97% time saving) - It is possible to model the interval uncertainties in pipeline parameters and the lower/upper bounds of the outputs.
		IEEE-123 distribution system and an 11-node natural gas system with 30 single-phase PV generators and an individual capacity of 15 kW.		

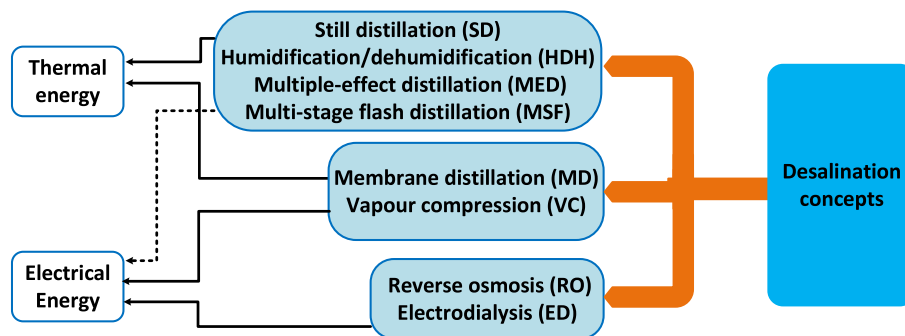


Fig. 5. Water desalination concepts.

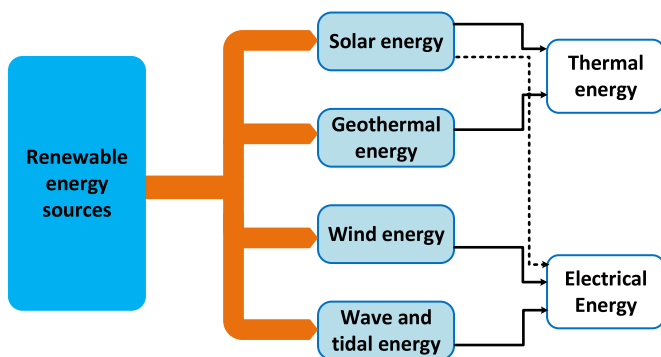


Fig. 6. Renewable energy sources.

the required energy and capital investments (Shannon et al., 2008). The reported statistics estimate that almost 8.78 Mt of oil are needed annually for generating 1 M/m³/day freshwater using a desalination system (Kalogirou, 2005), which proves that an alternative energy source is required for desalination systems. Researchers have made considerable efforts in the field of multi-carrier energy systems with the integration of water and electricity energy sources with the main objective of decreasing the total cost of the water desalination process. In (Novosel et al., 2015), the authors have accomplished a sensitivity analysis on the configuration of integrated power and water desalination systems with the penetration of renewable energy sources on CO₂ emissions and annual costs. In this reference, the analysis has shown that a 50%

decrease or increase of the fuel cost varied the total cost of the system by 29% and 48% annually in the usual scenario and high penetration of renewable energy resources, respectively. The authors have investigated the coupled power and water desalination systems in the islands of Lanzarote and Fuerteventura in (Nuez and Garcia Latorre, 2016), where an average of 167.01 €/MWh cost for power production in weak insular power networks is obtained. The authors in this reference have analyzed the flexibility in the water supply that is influential in the management of the power generation units considering the possibility of storage so that more than 11% can be saved in energy supply costs. The environmental effects of desalination as well as seawater desalination in the world have been studied in (Lattemann and Höpner, 2008), where methods of mitigating the influences of desalination on the environment are investigated. Solar energy is the best source among all the alternative energy sources taking into account the potential for supplying future energy requirements (Thiel et al., 2013). Recently, the remarkable effort has been made in the area of multi-carrier energy systems with the integration of water and electricity energy sources as defined as desalination systems. The main classification of desalination systems is based on desalination process, which include still desalination (SD), humidification/dehumidification (HDH), multiple-effect desalination (MED), multi-stage flash desalination (MSF), membrane desalination (MD), vapour compression (VC), reverse osmosis (RO), and electrodialysis (ED). The most worldwide accomplished desalination systems are RO, MED and MSF systems (Alkaisi et al., 2017). RO system takes advantage of low investment cost, maintenance and consuming energy and pollutant gas

emissions. A general overview of water desalination concepts is demonstrated in Fig. 5. In addition, Fig. 6 shows renewable energy sources can be utilized in the process of water desalination. The application of renewable energy sources for utilization in desalination systems has been introduced as a very promising selection especially in remote and arid regions (Koroneos et al., 2007).

A study has been carried out in the area of integrated electrical energy and seawater desalination units utilizing concentrated solar power technology in Oman (Gastli et al., 2010). Concentrating solar power (CSP) is studied in this research for generating both electrical energy and desalinated water, which is appropriately sited in the Duqum region considering the determination of solar radiation, geographical topology, and expansion plan of the project. The CSPs take advantage of no pollutant gas emissions, domestic, sufficiently large to meet the huge demand. The application of CSPs to desalination systems in the Middle East and North Africa (MENA) region is performed in (Center, 2007) for providing electrical energy and water. The analysis of the combination of CSPs and desalination systems is accomplished in terms of economic emission viewpoints for Egypt considering a framework for exporting the surplus provided electrical energy or water to Germany.

The authors have studied three systems utilizing an integration of reverse osmosis (RO) subsystem for generating water and a steam power plant for producing electrical energy in (Bouzayani

et al., 2009). The proposed model is investigated based on the nature of the power recovery unit the existence of thermal coupling among two subsystems considering feed water flow rate and salinity, energy recovery system, and operating pressures. The authors have proposed a new power-water combined plant (PWCP) founded based on low-grade waste heat to provide electrical energy and water in (He, W. et al., 2017b). The proposed model is based on integrating humidification dehumidification (HDH) desalination systems to the organic Rankine cycle (ORC), where the ORC system is taken into account as the top cycle of driving HDH system, which is capable to generate both freshwater and electrical energy. In (He, W. et al., 2017a), an integration of air-heated HDH desalination to ORC systems is introduced, where the organic fluid is evaporated by reheating the hot concentrated brine flowing out of the humidifier in order to generate electrical energy. The proposed model in this research is able to supply both electrical energy and water demands, which is capable to change mass flow rate ratio in order to supply different levels of power and water demand. A new hybrid photovoltaic technology is proposed in (Moh'd A and Al-Ammari, 2020) for generating power and water, which includes a solar-driven Stirling engine cooled by saline water. In (Zhang and Vesselinov, 2016), a bi-level programming problem is solved to minimize total electricity generation and water treatment cost subjecting to some operational constraints such as mass balance

Table 2

Studies in the area of water desalination systems with different objectives, contributions and case studies.

Reference	Main contributions	Type of desalination system	Case study	Major findings
Khan and Khan (2017)	Investigation of different desalination technologies for Karachi nuclear power plant proposing Viable techno-economic framework	RO, MED, MSF	Karachi nuclear power plant	The water cost of the RO + MED, MSF and MED almost equal to 1.25, 1.58 and 1 \$/m ³ , respectively.
Chafidz et al. (2014)	Development of a portable solar-based desalination system for autonomous application in arid remote areas	Vacuum Multi-Effect MD (V-MEMD)	A remote area in Saudi Arabia	The maximum value of fresh water production was 99.6 L/h under the conditions: •The heat pump utilization •The higher thermal-tank temperature •The low feed flow rate •The high solar radiance The average conductivity of the pure water and the optimum feed flow rate were 6.2 µs/cm and 69 L/h, respectively.
Hosseini et al. (2012)	Multi-objective designing of a combined gas turbine and multi-stage flash desalination plant considering energetic, economic, environmental and reliability viewpoints	Multi-stage flash desalination	A case study consisting of a gas turbine and desalination system	The cost of products and environmental cost impact are reduced by 13.4% and 53.4%, respectively. Moreover, the total exergy efficiency is increased by 14.8%.
Srithar et al. (2016)	Investigation of solar desalination system including triple basin glass solar still with cover cooling and parabolic dish concentrator	Triple basin solar still (TBSS)	A real stand-alone solar desalination system	Maximum fresh water of 16.94 kg/m ² ·day is achieved using the TBSS. The TBSS with charcoal and river sand enhances the distillate by 34.2 and 25.6%, respectively.
Al-Otoom and Al-Khalailah (2020)	Application of hygroscopic water solution and Kaoliang to improve the solar desalination	Humidification –dehumidification process using hygroscopic solutions and rotating belt	A continuous solar rotating belt desalination unit	The use of kaolin has increased the desalted water output from 10.8 kg/m ² ·day of using pure tap water to 32.1 kg/m ² ·day. The maximum efficiency is increased from 8.7 to 61% using kaolinite.
Alsarayreh et al. (2020)	Adding an energy recovery unit to RO process and economic assessment of two cases “with” and “without” energy recovery utilization	RO	A low salinity brackish water reverse osmosis system of Arab Potash Company	The total energy consumption is reduced between 47 and 53.8% while adding the energy recovery device.
(Nayar and Lienhard V, 2020)	RO, closed circuit RO, electrodialysis reversal (EDR) and monovalent selective EDR (MS-EDR) approaches have been compared in terms of cost and energy.	RO, closed circuit RO, electrodialysis reversal (EDR) and monovalent selective EDR (MS-EDR)	Ventura county	For <10-ha greenhouses, RO is the most cost-effective strategy. For >10-ha greenhouses: MS-EDR is more economic if it retain at least 20% of calcium and magnesium required for growing. The closed circuit RO is a good solution if the feedwater plus brine disposal cost is greater than 0.24\$/m ³ . If this cost reaches \$0.32/m ³ , more investment for closed circuit RO, EDR and MS-EDR, pays back in 2.4, 3.4 and 2.1 years .

and availability of fossil fuels, electricity balance criterion, power and water production capacities, energy demand for water treatment, water requirement of electricity generation process, availability of water resources including groundwater, surface and grey water, and permitted range of CO₂ emission production. Impacts of seasonal variability on optimal long-term operation of household water-energy-food microgrids is evaluated in (Wa'el A et al., 2018). It is proved that the use of grey water for industrial and agricultural applications and the anaerobic digestion of waste food and wastewater sludge for energy recovery reduces total potable water and electricity demand shortage, respectively. In (Jabari et al., 2020), a novel design of integrated cooling, potable water and power system by utilizing solar irradiance has been proposed that uses a solar dish Stirling engine for power and heat co-production. Table 2 summarizes the studies related to water desalination systems with different objectives, contributions and case studies.

3.2. Hydropower plants

Hydropower generation as renewable energy source is defined as practical concept of production of large quantities of electrical energy, which plays important role in network stability improvement and supporting the extension of other types of renewable energy sources such as photovoltaic system or wind turbines (Tassinari et al., 2016). Such technology has been defined as a clean and renewable source, which is effective in mitigating environmental issues of the electrical energy production industry (Mao et al., 2017). The reported statistics demonstrate that the hydropower generation units allocate 16% of the total electrical energy generation of the world (Ardizzon et al., 2014). Hydropower units can be categorized according to the production capacity including large, small, mini, and micro-hydro generation units. Table 3 provides the classification of hydropower units considering the generation capacity (Mohibullah et al., 2004). Fig. 7 shows statistics of small hydroelectric plants potential all around the world.

The other classification of hydropower units is based on the head of water available, which includes high, medium and low head units. Table 4 prepares the classification of hydropower units based on the water head and appropriate turbines for implementation in each category (Penche, 2004).

3.2.1. Power generation of hydro plants

Hydropower units convert the water pressure to mechanical power used to derive an electrical generator. The produced electrical energy is based on the pressure head and volume flow rate (Paish, 2002). The gross electrical energy and related hydropower can be obtained by using the water density, gravity, pressure head, water volume flow and time interval as follows (Purohit, 2008):

$$E_0 = \rho g Q H \Delta t \quad (16)$$

$$P_0 = \rho g Q H \quad (17)$$

Hand Q are the respective indicators of pressure head (m) and volume flow rate (m³/s). The water density (kg/m³) and gravity (m/s²) are defined by ρ and g .

Table 3
Classification of hydropower plants based on generation capacity.

Hydropower unit type	Capacity (kW)
Large	more than 1000 kW
Small	between 500 and 1000 kW
Mini	between 100 and 500 kW
Micro	less than 100 kW

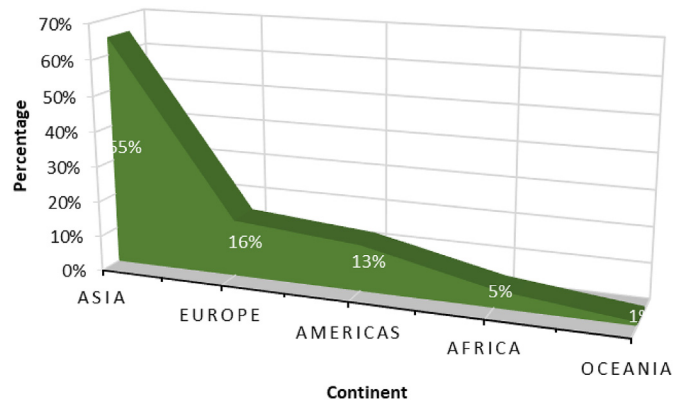


Fig. 7. Small hydroelectric plants potential in the world.

Table 4
Classification of hydropower plants based on water head.

Hydropower unit type	Head (m)
High head	more than 100 m
Medium head	between 30 and 100 m
Low head	between 2 and 30 m

The available electrical energy and related hydropower are based on the total efficiency of the hydropower system. The ratio of useful energy to total expended energy is defined as the efficiency of a system. The efficiency of a hydropower system consists of hydraulic and mechanical efficiency. The ratio of electrical energy provided by the runner of a turbine to electrical energy provided at the inlet of a turbine is called hydraulic efficiency (η_h), which can be stated as:

$$\eta_h = \frac{P_R}{P_W} \quad (18a)$$

Mechanical efficiency (η_m) is defined as available power at shaft per runner power, which can be stated as follows:

$$\eta_m = \frac{P_S}{P_R} \quad (18b)$$

The available hydropower can be attained considering the total efficiency of the system (η) as follows:

$$\eta = \eta_h \eta_m \quad (19)$$

$$E = \eta E_0 \quad (20)$$

$$P = \eta P_0 \quad (21)$$

Thermal power generation units require high operational costs; however, the initial development costs of such plants are lower. Moreover, the operation cost of hydropower units is ignorable; however, such units need high construction costs. Accordingly, the integration of hydro and thermal plants is defined as a practical solution for supplying power demand due to economic and technical viewpoints (Nadakuditi et al., 2016).

The largest hydropower plant in the world is the Three Gorges Reservoir in China located in Yichang, Hubei province, which has a total installed capacity of 22,500 MW (Huang et al., 2015). The second-largest hydropower plant in the world is the Itaipu power plant located on the Parana River, which has an installed capacity of 14,000 MW (Mariano et al., 2016). The third and fourth-largest

hydropower plant in the world is the Guri located in Venezuela (Matejcek, 2017) and Tucuruí located in Brazil (Oliveira and Nogueira, 2016). The respective installed capacity of the Guri and Tucuruí hydropower plants is 10,200 MW and 8370 MW. The fifth to tenth largest hydropower plants in the world are Grand Coulee located in United States of America, Sayano-Shushenskaya located in Russia, Longtan in China, Krasnoyarsk in Russia, Robert-Bourassa and Churchill Falls in Canada.

3.2.2. Short-term hydrothermal scheduling

The main objective of the short-term hydrothermal scheduling (STHS) problem is to obtain minimum operation cost of hydro plants and thermal units to meet the daily power load demand of the power system. It is notable that the daily operation cost of hydro plants is negligible, which is not considered in the objective function of the STHS problem. Consequently, the objective function of the STHS problem is the operation cost of thermal plants to satisfy power demand as (21).

$$P_{Load}^t = \sum_{i=1}^{N_p} P_i^t + \sum_{k=1}^{N_h} P_k^t \quad (22)$$

where the number of thermal plants and hydro units is indicated by N_p and N_h . Moreover, the power load demand at each time interval is defined by P_{Load}^t . The objective function and constraints of the STHS problem can be stated as follows:

$$\min \sum_{t=1}^{24} \sum_{i=1}^{N_p} C_i(P_i^t) \quad (23)$$

The power supply of hydro plants is based on water release of the hydro reservoir and its volume, which can be modeled as:

$$P_k^t = C_{1k}(V_k^t)^2 + C_{2k}(Q_k^t)^2 + C_{3k}(V_k^t Q_k^t) + C_{4k}(V_k^t) + C_{5k}(Q_k^t) + C_{6k} \quad (24)$$

where the power supply of the k th hydro plant at time t is defined by P_k^t . Water release of the k th hydro reservoir and its volume at time t are indicated by Q_k^t and V_k^t , respectively. The power generation coefficients of the k th hydro unit are indicated by C_{1k} , C_{2k} , C_{3k} , C_{4k} , C_{5k} and C_{6k} .

The power supply of each hydro plant at each time should be limited to its lower and upper bounds as:

$$P_k^{\min} \leq P_k^t \leq P_k^{\max} \quad (25)$$

where, $P_k^{\min}$ and $P_k^{\max}$ are the respective minimum and maximum power production of the k th hydro unit.

It should be noted that water release and reservoir volume of each hydro plant have an operational limitation to consider their acceptable bounds, which can be stated as:

$$V_k^{\min} \leq V_k^t \leq V_k^{\max} \quad (26)$$

$$Q_k^{\min} \leq Q_k^t \leq Q_k^{\max} \quad (27)$$

where, $V_k^{\min}$ and $V_k^{\max}$ are the respective minimum and maximum reservoir volume of the k th hydro unit. Moreover, $Q_k^{\min}$ and $Q_k^{\max}$ indicate the lower and upper bounds of the water release of the k th hydro plant.

The dynamic water balance in each reservoir of the hydro unit considers a balance among reservoir storage volume at each time and its previous time, inflow rate, water discharge and spillage, which can be stated as follows:

$$V_k^t = V_k^{t-1} + I_k^t - Q_k^t - S_k^t + \sum_{m \in R_{up}^k} (Q_m^{t-Td_m} + S_m^{t-Td_m}) \quad (28)$$

where, V_k^t and V_k^{t-1} define water storage volume of the k th hydro plant at times t and $t-1$, respectively. In addition, I_k^t and S_k^t are the respective indicators of inflow rate and spillage of the k th hydro plant. Finally, it is usually assumed in the daily operation of hydro units that the initial and final water volume of each hydro reservoir are determined, which can be modeled as:

$$V_k^0 = V_k^{init} \quad (29)$$

$$V_k^{24} = V_k^{end} \quad (30)$$

where the initial and final water volume of the k th hydro plant is indicated by V_k^{init} and V_k^{end} . Moreover, V_k^0 and V_k^{24} are the respective indicators of the water volume of the k th hydro plant at time intervals 0 and 24. The operation cost of conventional thermal plants is modeled as the sum of a quadratic function of supplied power and a sinusoidal term to the model valve-point influence of such plants, which can be formulated as follows:

$$C_i^t(P_i^t) = a_i(P_i^t)^2 + b_i P_i^t + c_i + |e_i \sin(f_i(P_i^{\min} - P_i^h))| \quad (31)$$

where, $C_i^t(P_i^t)$ and P_i^t are the operation cost and power supplied by the i th thermal unit at time t . The cost coefficients of the i th thermal unit are defined by a_i , b_i , c_i , e_i and f_i . The power production of each thermal unit at each time interval should be limited to its lower and upper bounds as follows:

$$P_i^{\min} \leq P_i^t \leq P_i^{\max} \quad (32)$$

where, $P_i^{\min}$ and $P_i^{\max}$ are the respective indicators of lower and upper power production bounds of the i th thermal.

3.3. Scheduling and planning of hydro-based power plants

The optimal generation scheduling and planning of hydropower sources is considered an essential issue of electric energy systems. Such an issue has been studied as generation scheduling of hydro plants, hydrothermal units, pumped storage units or a combination of such units with other power sources.

3.3.1. Scheduling and planning of hydro units

The main objective of hydro plants scheduling is obtaining optimal schedules of power supply for either an individual hydro unit or a combined network of cascaded watercourses in a determined time of one day of a one week (Kong et al., 2020). The optimal generation scheduling of hydropower plants should be studied considering several electrical and operational constraints, which include complex hydraulic coupling of hydro plants, limitations of water discharge, water storage, power production of hydro units, power load demand of the system, and water balance of the hydro plants (Warland et al., 2016). The integration of hydropower and renewable energy sources such as photovoltaic (PV) power has been defined as an effective rule in providing the power load demand of a determined region, which has been investigated in (Yang

et al., 2018). In this reference, a stochastic optimization is proposed for long-term operating rules for such an integrated system using considering uncertainties in reservoir inflow and PV power. Several efforts have been concentrated on short-term and long-term scheduling of hydro units considering different models. A unit commitment problem of the hydro plants is studied by considering mid/long-term planning and an approximation of operation cost function of such plants in (Guedes et al., 2017), which has shown the positive impact of the proposed model on decreasing the operation cost of Brazilian hydro-power units. A scheduling framework for hydro-PV-wind operation management by taking firm generation limits into account is proposed in (Liu et al., 2016), where the numerical analysis on a realistic hydro system in Yunnan has demonstrated the capability of the proposed model in dealing with uncertain nature of wind power. The authors in (Irshad et al., 2016) have studied the integration of small hydro units with compress air storage systems to overcome the lack of considerable storage capability of micropower plants in a system located in Dor River in Pakistan. A model for weekly scheduling of hydro plants in energy and reserve markets is introduced in (Chazarra et al., 2016), which was able to handle the uncertainty in real-time operation of the energy systems considering regulation-up and down reserves. The authors have studied the scheduling of hydro units considering uncertainties in (Yuan et al., 2016, 2017) using random fuzzy programming for hydropower plants located in Qing river in China, which demonstrated the ability of the introduced model in avoiding frequent plant switches to satisfy uncertain power demand. In (Castronuovo et al., 2016), the scheduling of hydro units in electricity markets considering the influence of transmission constraints and zonal prices on the optimal management of plants have been discussed, which has shown the significant effect of such consideration in the positioning of hydro units. The authors have focused on analysis of greenhouse gas emissions reduction in hydropower industry of five largest hydro-based projects in China in (Li et al., 2017), which has proven that a significant decrement of gas emissions is attained by pre-impoundment clearance, which is known as engineering measurement of removal of vegetation cover, buildings, structures and solid waste landfill as a clean method for production in the hydropower sector. A new heuristic-based optimization method for determining optimal set points of hydro plants has been proposed in (Skjelbred et al., 2020) concerning influence of time-varying head, losses associated with intake, penstock, and tailrace, and efficiency of turbine.

3.3.2. Scheduling and planning of hydrothermal units

Thermal power production plants need high operational costs; however, the initial development costs of such plants are lower. In addition, the operation cost of hydropower units is ignorable; however, such units need high construction costs. Accordingly, the integration of hydro and thermal plants is defined as a practical solution for supplying power demand due to economic and technical viewpoints (Fang et al., 2014). The optimal generation scheduling of hydrothermal systems obtains optimal scheduling of both hydro and thermal plants, simultaneously considering minimum operation cost of the system. Several mathematical and heuristic optimization methods have been applied for hydrothermal scheduling such as non-linear programming (NLP) (Catalão et al., 2011), OptQuest/NLP (OQNLP) method (Hoseynpour et al., 2017) and harmony search (Nazari-Heris et al., 2018a). The authors have studied the integration of hydrothermal plants and wind power as an environment friendly and clean energy to minimize the operation cost and emission of pollutant gases by thermal plants in (Dasgupta et al., 2020). In this reference, optimal power flow constraints is considered when optimizing the system optimal

operation, which is tested on a standard 9-bus power network. In addition, various contributions have been proposed in recent publications, which include multi-objective economic emission dispatch of hydrothermal systems (Feng et al., 2017), self-scheduling of hydro-thermal systems (Soroudi, 2013), the solution of hydrothermal scheduling considering load uncertainty issue (Charwand et al., 2016), long-term planning of hydrothermal systems considering risk-averse policies (Larroyd et al., 2017), participation of hydrothermal generation companies (GENCOs) in power market (Padmini and Jegatheesan, 2016), stochastic short-term hydrothermal scheduling (Bezerra et al., 2017), and stochastic midterm financial risk constrained (Wu et al., 2008). The authors have proposed grey wolf optimization method for dealing with a multi-objective scheduling of hydrothermal plants in (Pradhan et al., 2020), where the equality and inequality constraints of valve-point loading, water discharge, and water storage are concerned. To provide an influential review on optimal scheduling of hydrothermal systems, comparative researches with different objective functions and constraints are summarized in Table 5.

3.3.3. Scheduling and planning of pumped storage units

Pumped storage units are considered as commercial large-scale energy storage technology considering more than 300 installed units in the world with a total installed capacity of over 95 GW (Roberts, 2009). Recently, remarkable interests have been noticed in the area of pumped storage construction all over the world such as the capacity of 36 GW electrical energy for European Union, the capacity of 21.8 GW for the USA, and capacity of 24.5 GW for Japan (Deane et al., 2010). Different contributions have been made in recent publications in the area of pumped storage units. A scheduling model for an isolated power system consisting of pumped storage unit and wind power is proposed in (Pérez-Díaz and Jiménez, 2016), which has demonstrated that the presence of pumped storage units are effective in decreasing the operation cost of the whole integrated power and water systems by an almost 2.5–11% rate. In addition, the application of such units is lucrative in integrating wind power to the hybrid system that is effective in dispensing with some inflexible thermal power plants. The authors have proposed different models for bidding strategies of pumped storage units in power markets in (Kanakasabapathy and Swarup, 2010; Lu et al., 2003), where a multistage-looping model is applied for meeting the reservoir capacity constraints within each time segment that are used to develop a fixed-schedule bidding strategy for a season. Design of pumped-storage wind system, economic investigation of pumped storage and electrical energy storage systems, and load shedding frameworks for power systems using pumped-storage units.

4. Multi-carrier energy systems with integration of electricity, water and gas sources

The application of natural gas (NG) and liquefied natural gas (LNG) for generating electrical energy as an alternative source of grid electricity at desalination systems has been studied in (Cherchi et al., 2017). The life cost analysis has been accomplished for such an idea considering cost and environmental viewpoints of different power supply choices for desalination systems. The analysis demonstrated that such a concept for generating electrical energy using NG and LNG for the desalination system is more economical and environmentally friendly with respect to the direct power buy from the grid. Accordingly, the development of gas-fired power generation plants alongside water infrastructures should be accomplished considering several issues such as customer requirements, factors related to the site, parameters related to design and operating conditions. The cost structure of a combination of

Table 5

Studies in the area of hydrothermal systems with different objectives, contributions and case studies.

Reference	Proposed approach and contributions	Test system	Advantages and disadvantages of the proposed models
Feng et al. (2017)	Minimizing the operation cost of hydrothermal systems as well as the emission of pollutant gases by thermal plants to meet daily load demand using a multi-objective genetic algorithm	4 cascaded hydro plants and 3 thermal plants	The advantage is dealing with two conflicting objectives in the optimal operation of hydrothermal systems including cost and pollutant gas emissions; however, the main disadvantage is not considering the power and water network constraints as well as power transmission losses.
Soroudi (2013)	Self-scheduling of hydrothermal scheduling considering uncertainties of power market price using a robust optimization method	4 cascaded hydro plants and 11 thermal plants	Although the capability of finding robust optimal scheduling of hydro and thermal plants with respect to the uncertain power market price is the main advantage of the model, the model suffers from a network-constrained model and pollutant gas emissions from traditional thermal plants.
Charwand et al. (2016)	Optimal energy management of hydrothermal systems in the presence of uncertainty associated with power load demand applying information gap decision theory	• 4 cascaded hydro plants and 3 thermal plants	The model has the ability to obtain a robust solution for energy management of the hydrothermal system with respect to the uncertain load demands; however, it is not successful to consider the uncertainty of power market price.
Haghrab et al. (2014)	• Employment of real-coded genetic algorithm to obtain optimal scheduling of hydrothermal systems considering influences of valve point, power flow loss	• 4 cascaded hydro plants and 1 thermal plant • 4 cascaded hydro plants and 3 thermal plants	The main advantage of this study is obtaining the minimum operation cost for the first hydrothermal system with respect to all works in this area and the proposed scheduling model has been solved in a deregulated environment; however, it suffers from considering power and water systems as well as the uncertainty of power demand.
Roy et al. (2013)	• Implementation of teaching learning-based optimization method to determine optimal scheduling of hydrothermal systems considering prohibited operating zones of hydro units and influences of valve loading	• 4 cascaded hydro plants and 1 thermal plant • 4 cascaded hydro plants and 3 thermal plants	Attaining insignificant calculation time in the optimization process and high convergence speed, robust characteristics of the introduced optimization approach are the major advantages of the study; however, no network constraint has been considered for the power and water system, and smart grid characteristics in terms of power exchange with the market have not been addressed.
Zhang et al. (2013)	• The solution of multi-objective scheduling of hydrothermal systems with consideration of valve influences and pollutant gas emissions using differential evolution	• 4 cascaded hydro plants and 3 thermal plants	Although the introduced approach in this study has the capability of handling economic-environmental issues of hydrothermal systems, it is not successful to address the power flow and constraints of the networks.
Basu (2014)	• Application of differential evolution in attaining the optimal management of hydrothermal systems taking into account valve point influences, power losses, prohibited operating zones of water reservoirs and ramp rates limits	• 4 cascaded hydro plants and 1 thermal plant • 4 cascaded hydro plants and 3 thermal plants • 4 hydro plants and 10 thermal plants	Attaining less calculation time in optimization and robust characteristics of the applied optimization method; however, no discussion on the power exchange with the upper grid is done.
Nazari-Heris et al. (2018b)	• Application of robust optimization method to study the scheduling problem of hydrothermal systems in the presence of uncertain parameters	• 4 cascaded hydro plants and 1 thermal plant	Attaining robust scheduling of hydro and thermal plants against power market uncertainty and considering the deregulated environment; however, network constraints are not taken into account.
López-Salgado et al. (2020)	Stochastic short-term scheduling of cascaded hydrothermal units with controllable reservoirs using Benders decomposition	A real system of 10 thermal plants, 2 wind farms and 11 hydroelectric stations	Computational efficiency of Benders decomposition for optimal scheduling of large-scale hydrothermal systems
Rasku et al. (2020)	Hydro-thermal scheduling and expansion planning considering the uncertain generation of the renewable energy resources	Nordic power system	Reducing the total yearly operation costs by 0.18–0.41%, the spillage of run-of-river hydropower and the curtailment of wind power by 0.42–0.47 and 0.05–0.07% points, respectively.
Franz et al. (2020)	Long-term coordination of hydro-thermal units for cost and emission control	A 10-plant system is exactly replicated to make a 20,30, ...,100 unit network.	Uncertainty of system configuration and demand as well as possible transmission expansions have not been considered in long-term studies.

utilities as firms operating in integrated electricity, gas and water sectors has been investigated in (Fraquelli et al., 2004). Composite cost function strategy proposed by Pulley and Braunstein, which is appropriate for multi-product firms, has been implemented in this research in order to study the cost properties of Italian public utilities with the integration of electricity, gas, and water. The obtained results show that small specialized firms will attain benefit increment by evolving in multi-utilities, which participate in similar sectors such as electricity, gas and water. On the other hand, positive impact on larger-scale utilities is not verified. In (Avritzer et al., 2015), three concepts for modeling and investigating survivability of smart infrastructures related to electricity, gas, and water, which is defined as recovery ability of services after the disaster occurrence. The first approach is based on the behavioral decomposition method, which models the process separately from the system performance for the electricity and gas system. In addition, a single integrated modeling method is employed for

dealing with the failure process of the water system integrating the failure and the water transportation and storage. The integration of various energy networks under the concept of energy systems of so-called multi-carrier energy systems can provide sufficient flexibilities to supply different types of energy demands like electricity, heat, gas and water demands. Optimal operation of multi-carrier energy systems including integrated electricity, gas and water networks has been studied subject to economic and environmental criteria in (Ghahramani et al., 2018; Majidi et al., 2017). In detail, multi-objective optimization frameworks have been proposed for the optimal economic-environmental operation of multi-carrier energy systems and weighted-sum and epsilon-constraint methodologies have been employed to solve the proposed models. Obtaining the optimal Pareto fronts via the mentioned methods, the fuzzy satisfying method has been employed to select the best trade-off solutions with/without demand response programs. In other studies, about multi-carrier energy systems, the uncertainty-

based performance of integrated electricity, heat, gas and water networks has been studied subject to the uncertainty of electricity price in (Majidi et al., 2019a, b). To model the uncertainty of hourly prices, robust optimization method and information gap decision theory have been respectively employed and the appropriate risk-averse/risk-neutral/risk-seeker operating strategies within the uncertainty are obtained. In (Ferdowsi et al., 2017), the authors have presented a unified scheme for integrated gas-power-water infrastructure, for which a novel game-theoretic scheme is employed for studying the security of this model. In this reference, the authors have formulated a zero-sum noncooperative game between an attacker seeking to revise the states of system for increasing the cost of the system and a defender allocating communication resources on detection of attack for system monitoring.

5. Conclusion and future trends

The multi-carrier energy systems with the integration of electricity, gas, and water energy sources, which are becoming more automated, have been introduced as up-to-date issues in terms of economic and environmental viewpoints. The statistics reported on the penetration of interconnecting elements such as gas-fired power plants, combined heat and power technology, water desalination systems, hydropower turbines and power to gas technology verified the high rate of interconnection among energy systems. In this study, a comprehensive survey has been provided for investigating the current literature on three classifications of multi-carrier energy systems, which include a) electricity and gas, b) electricity and water, and c) electricity, gas, and water energy sources. Each of the above-mentioned categories is introduced, and the main contributions of literature are provided according to the key elements and novelties. Moreover, the implementation of practical projects around the world in the area of integrated systems is introduced. Integrated electricity, gas and water energy sources as multi-carrier energy systems take several remarkable advantages. Such benefits include global, economic and environmental points such as the need of gas-fired generation plants to lower carbon footprint and fuel cost, and fast response of such generators to fluctuations of renewables in comparison with traditional coal-fired units. Also, integrated energy systems can produce useable water by desalination systems and attain electricity by hydroelectric generation units, which are introduced as the main reasons to concentrate on integrated electricity, gas and water-based energy systems. The investigated challenges and issues can largely be helpful for researchers studying in the area of combined electricity, gas and water systems. The analysis of the studied works in this paper shows that the co-optimization of the power, gas, heating and water systems provides the capability of lower operation and planning cost of the systems as well as augmentation viewpoint on the operation and planning of the systems. Additionally, the reports in the literature verify that the ability to convert different types of energy to each other in the integrated energy networks increases the efficiency and reliability of the integrated systems in terms of operation and expansion planning challenges.

As a future trend, optimal design and analyses of food-water-gas-power-environment nexus models should be studied by scholars to minimize their operation and maintenance costs. In addition, energetic, exergetic, economic and environmental (4E) benefits of heat/gas/power storage mediums or systems in multi-carrier hub systems should be reviewed. Moreover, optimal load flow analyses and demand-side management strategies, which are implemented to power systems, should also be applied to other demands such as gas and water. In fossil-fuels based generation infrastructures, capturing, transporting, and securely storing CO₂ emissions in abandoned gas and oil wells and recycling it by gas-

fired power generation systems, thermally-driven desalination units, and gas-load supplying process, should also be considered. The risk analysis of the integrated power, gas, heating and water systems is of great importance to be investigated in future works, where the effect of load uncertainty in one of the energy networks on the other systems schedule can be studied. Considering the penetration of energy storage technologies in real-world applications, the studies of multi-carrier energy storage systems can highlight the role of co-optimization of integrated energy systems in managing energy loads and increasing the performance of systems operation.

Declaration of competing interest

None.

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References

- Al-Mutaz, I.S., Wazeer, I., 2014. Comparative performance evaluation of conventional multi-effect evaporation desalination processes. *Appl. Therm. Eng.* 73 (1), 1194–1203.
- Al-Otoom, A., Al-Khalailah, A.T., 2020. Water desalination using solar continuous humidification–dehumidification process using hygroscopic solutions and rotating belt. *Sol. Energy* 197, 38–49.
- Al Ghamdi, A., Mustafa, I., 2016. Exergy analysis of a MSF desalination plant in Yanbu, Saudi Arabia. *Desalination* 399, 148–158.
- Alabdulwahab, A., Abusorrah, A., Zhang, X., Shahidehpour, M., 2015. Coordination of interdependent natural gas and electricity infrastructures for firming the variability of wind energy in stochastic day-ahead scheduling. *IEEE Trans. Sustain. Energy* 6 (2), 606–615.
- Alkai, A., Mossad, R., Sharifian-Barforoush, A., 2017. A review of the water desalination systems integrated with renewable energy. *Energy Procedia* 110, 268–274.
- Alsarayreh, A.A., Al-Obaidi, M.A., Al-Hroub, A.M., Patel, R., Mujtaba, I.M., 2020. Evaluation and minimisation of energy consumption in a medium-scale reverse osmosis brackish water desalination plant. *J. Clean. Prod.* 248, 119220.
- An, S., Li, Q., Gedra, T.W., 2003. In: *Natural Gas and Electricity Optimal Power Flow, Transmission and Distribution Conference and Exposition*, 2003 IEEE PES. IEEE, pp. 138–143.
- Ardizzon, G., Cavazzini, G., Pavesi, G., 2014. A new generation of small hydro and pumped-hydro power plants: advances and future challenges. *Renew. Sustain. Energy Rev.* 31, 746–761.
- Avritzer, A., Carnevali, L., Ghasemieh, H., Happe, L., Haverkort, B.R., Koziolk, A., Menasche, D., Remke, A., Sarvestani, S.S., Vicario, E., 2015. Survivability evaluation of gas, water and electricity infrastructures. *Electron. Notes Theor. Comput. Sci.* 310, 5–25.
- Bai, L., Li, F., Cui, H., Jiang, T., Sun, H., Zhu, J., 2016. Interval optimization based operating strategy for gas-electricity integrated energy systems considering demand response and wind uncertainty. *Appl. Energy* 167, 270–279.
- Bao, S., Yang, Z., Yu, J., Dai, W., Guo, L., Yu, H., 2019. Probabilistic energy flow and risk assessment of Electricity–Gas systems considering the thermodynamic process. *Energy* 189, 116263.
- Basu, M., 2014. Improved differential evolution for short-term hydrothermal scheduling. *Int. J. Electr. Power Energy Syst.* 58, 91–100.
- Bezerra, B., Veiga, A., Barroso, L.A., Pereira, M., 2017. Stochastic long-term hydrothermal scheduling with parameter uncertainty in autoregressive streamflow models. *IEEE Trans. Power Syst.* 32 (2), 999–1006.
- Bouzayani, N., Galanis, N., Orfi, J., 2009. Thermodynamic analysis of combined electric power generation and water desalination plants. *Appl. Therm. Eng.* 29 (4), 624–633.
- Castronuovo, E.D., Hermida, G., Gholami, M., Bovo, C., Berizzi, A., 2016. Optimal scheduling of a hydro basin in a pool-based electricity market with consideration of transmission constraints. *Elect. Power Syst. Res.* 131, 255–263.
- Catalão, J.P.d.S., Pousinho, H.M.L., Mendes, V.M.F., 2011. Hydro energy systems management in Portugal: profit-based evaluation of a mixed-integer nonlinear approach. *Energy* 36 (1), 500–507.
- Center, B.P., 2020. *Annual Energy Outlook 2020*.
- Center, G.A., 2007. *Concentrating Solar Power for Seawater Desalination*. Final Report, German Aerospace Center (DLR), Stuttgart.
- Chafidz, A., Al-Zahrani, S., Al-Otaibi, M.N., Hoong, C.F., Lai, T.F., Prabu, M., 2014. Portable and integrated solar-driven desalination system using membrane

- distillation for arid remote areas in Saudi Arabia. *Desalination* 345, 36–49.
- Chandrashekhara, M., Yadav, A., 2017. Water desalination system using solar heat: a review. *Renew. Sustain. Energy Rev.* 67, 1308–1330.
- Charwand, M., Ahmadi, A., Sharaf, A.M., Gitizadeh, M., Esmaeel Nezhad, A., 2016. Robust hydrothermal scheduling under load uncertainty using information gap decision theory. *Int. Trans. Electr. Energy Syst.* 26 (3), 464–485.
- Chaudry, M., Jenkins, N., Qadrdan, M., Wu, J., 2014. Combined gas and electricity network expansion planning. *Appl. Energy* 113, 1171–1187.
- Chazarra, M., García-González, J., Pérez-Díaz, J.I., Arteseros, M., 2016. Stochastic optimization model for the weekly scheduling of a hydropower system in day-ahead and secondary regulation reserve markets. *Elec. Power Syst. Res.* 130, 67–77.
- Chen, S., Wei, Z., Sun, G., Cheung, K.W., Wang, D., 2017. Identifying optimal energy flow solvability in electricity-gas integrated energy systems. *IEEE Trans. Sustain. Energy* 8 (2), 846–854.
- Cheng, L., Liu, C., Huang, R., Li, H., 2016. An optimal operating strategy for CCHP in multi-energy carrier system. In: 2016 IEEE Power and Energy Society General Meeting (PESGM). IEEE, pp. 1–5.
- Cherchi, C., Badruzzaman, M., Becker, L., Jacangelo, J.G., 2017. Natural gas and grid electricity for seawater desalination: an economic and environmental life-cycle comparison. *Desalination* 414, 89–97.
- Cominola, A., Spang, E., Giuliani, M., Castelletti, A., Lund, J., Loge, F., 2018. Segmentation analysis of residential water-electricity demand for customized demand-side management programs. *J. Clean. Prod.* 172, 1607–1619.
- Cui, H., Li, F., Hu, Q., Bai, L., Fang, X., 2016. Day-ahead coordinated operation of utility-scale electricity and natural gas networks considering demand response based virtual power plants. *Appl. Energy* 176, 183–195.
- Dasgupta, K., Roy, P.K., Mukherjee, V., 2020. Power flow based hydro-thermal-wind scheduling of hybrid power system using sine cosine algorithm. *Elec. Power Syst. Res.* 178, 106018.
- Deane, J., Ciaráin, M.O., Gallachóir, B.O., 2017. An integrated gas and electricity model of the EU energy system to examine supply interruptions. *Appl. Energy* 193, 479–490.
- Deane, J.P., Gallachóir, B.O., McKeogh, E., 2010. Techno-economic review of existing and new pumped hydro energy storage plant. *Renew. Sustain. Energy Rev.* 14 (4), 1293–1302.
- Diagoupis, T.D., Andrianesis, P.E., Dialynas, E.N., 2016. A planning approach for reducing the impact of natural gas network on electricity markets. *Appl. Energy* 175, 189–198.
- Eissa, M.M., 2018. First time real time incentive demand response program in smart grid with “i-Energy” management system with different resources. *Appl. Energy* 212, 607–621.
- Elaiw, A., Xia, X., Shehata, A., 2012. Solving dynamic economic emission dispatch problem with valve-point effects using hybrid DE-SQP. In: Power Engineering Society Conference and Exposition in Africa (PowerAfrica), 2012 IEEE. IEEE, pp. 1–6.
- Elaiw, A., Xia, X., Shehata, A., 2013. Hybrid DE-SQP and hybrid PSO-SQP methods for solving dynamic economic emission dispatch problem with valve-point effects. *Elec. Power Syst. Res.* 103, 192–200.
- Erdener, B.C., Pambour, K.A., Lavin, R.B., Dengiz, B., 2014. An integrated simulation model for analysing electricity and gas systems. *Int. J. Electr. Power Energy Syst.* 61, 410–420.
- Fang, N., Zhou, J., Zhang, R., Liu, Y., Zhang, Y., 2014. A hybrid of real coded genetic algorithm and artificial fish swarm algorithm for short-term optimal hydro-thermal scheduling. *Int. J. Electr. Power Energy Syst.* 62, 617–629.
- Fathima, A.H., Palanisamy, K., 2015. Optimization in microgrids with hybrid energy systems—A review. *Renew. Sustain. Energy Rev.* 45, 431–446.
- Feng, Z.-K., Niu, W.-J., Zhou, J.-Z., Cheng, C.-T., Qin, H., Jiang, Z.-Q., 2017. Parallel multi-objective genetic algorithm for short-term economic environmental hydrothermal scheduling. *Energies* 10 (2), 163.
- Ferdowsi, A., Sanjab, A., Saad, W., Mandayam, N.B., 2017. Game Theory for Secure Critical Interdependent Gas-Power-Water Infrastructure, 2017 Resilience Week (RWS). IEEE, pp. 184–190.
- Franz, A., Rieck, J., Zimmermann, J., 2020. A long-term unit commitment problem with hydrothermal coordination for economic and emission control in large-scale electricity systems. *OR Spectrum* 42 (1), 235–259.
- Fraquelli, G., Piacenza, M., Vannoni*, D., 2004. Scope and scale economies in multi-utilities: evidence from gas, water and electricity combinations. *Appl. Econ.* 36 (18), 2045–2057.
- Gastli, A., Charabi, Y., Zekri, S., 2010. GIS-based assessment of combined CSP electric power and seawater desalination plant for Duqum—Oman. *Renew. Sustain. Energy Rev.* 14 (2), 821–827.
- Gazijahani, F.S., Salehi, J., 2018. Reliability constrained two-stage optimization of multiple renewable-based microgrids incorporating critical energy peak pricing demand response program using robust optimization approach. *Energy* 161, 999–1015.
- Ghahramani, M., Nojavan, S., Zare, K., Mohammadi-ivatloo, B., 2018. Application of Load Shifting Programs in Next Day Operation of Distribution Networks, Operation of Distributed Energy Resources in Smart Distribution Networks. Elsevier, pp. 161–177.
- Gil, M., Dueñas, P., Reneses, J., 2016. Electricity and natural gas interdependency: comparison of two methodologies for coupling large market models within the European regulatory framework. *IEEE Trans. Power Syst.* 31 (1), 361–369.
- Guedes, L.S., de Mendonça Maia, P., Lisboa, A.C., Vieira, D.A.G., Saldanha, R.R., 2017. A unit commitment algorithm and a compact MILP model for short-term hydro-power generation scheduling. *IEEE Trans. Power Syst.* 32 (5), 3381–3390.
- Haghragh, A., Mohammadi-ivatloo, B., Seyedmonir, S., 2014. Real coded genetic algorithm approach with random transfer vectors-based mutation for short-term hydro-thermal scheduling. *IET Gener., Transm. Distrib.* 9 (1), 75–89.
- He, C., Wu, L., Liu, T., Shahidehpour, M., 2017. Robust co-optimization scheduling of electricity and natural gas systems via ADMM. *IEEE Trans. Sustain. Energy* 8 (2), 658–670.
- He, W., Zhang, X., Han, D., Gao, L., 2017a. Performance analysis of a water-power combined system with air-heated humidification dehumidification process. *Energy* 130, 218–227.
- He, W., Zhu, W., Han, D., Huang, L., Wu, Y., Zhang, X., 2017b. Performance simulation of a power-water combined plant driven by low grade waste heat. *Energy Convers. Manag.* 145, 107–116.
- Hoseynpour, O., Mohammadi-ivatloo, B., Nazari-Heris, M., Asadi, S., 2017. Application of dynamic non-linear programming technique to non-convex short-term hydrothermal scheduling problem. *Energies* 10 (9), 1440.
- Hosseini Imani, M., Niknejad, P., Barzegaran, M.R., 2019. Implementing Time-of-Use Demand Response Program in microgrid considering energy storage unit participation and different capacities of installed wind power. *Elec. Power Syst. Res.* 175, 105916.
- Hosseini, S.R., Amidpour, M., Shakib, S.E., 2012. Cost optimization of a combined power and water desalination plant with exergetic, environment and reliability consideration. *Desalination* 285, 123–130.
- Hu, X., Zhao, X., Feng, X., 2019. Probabilistic-interval energy flow analysis of regional integrated electricity and gas system considering multiple uncertainties and correlations. *IEEE Access* 7, 178209–178223.
- Huang, L., Fang, H., Reible, D., 2015. Mathematical model for interactions and transport of phosphorus and sediment in the Three Gorges Reservoir. *Water Res.* 85, 393–403.
- Irshad, U.B., Javadi, M., Shafiq, S., Alam, M.S., Abido, M., Mumtaz, T., 2016. Optimization of daily operation of micro hydro power plant coupled with compress air storage. In: Multi-Topic Conference (INMIC), 2016 19th International. IEEE, pp. 1–5.
- Jabari, F., Jabari, H., Mohammadi-ivatloo, B., Ghafouri, J., 2019. Optimal short-term coordination of water-heat-power nexus incorporating plug-in electric vehicles and real-time demand response programs. *Energy* 174, 708–723.
- Jabari, F., Nazari-heris, M., Mohammadi-ivatloo, B., Asadi, S., Abapour, M., 2020. A solar dish Stirling engine combined humidification-dehumidification desalination cycle for cleaner production of cool, pure water, and power in hot and humid regions. *Sustain. Energy Technol. Assess.* 37, 100642.
- Jabari, F., Nojavan, S., Ivatloo, B.M., Shariifan, M.B., 2016. Optimal short-term scheduling of a novel tri-generation system in the presence of demand response programs and battery storage system. *Energy Convers. Manag.* 122, 95–108.
- Kahrl, F., Hu, J., Kwok, G., Williams, J.H., 2013. Strategies for expanding natural gas-fired electricity generation in China: economics and policy. *Energy Strat. Rev.* 2 (2), 182–189.
- Kalogirou, S.A., 2005. Seawater desalination using renewable energy sources. *Prog. Energy Combust. Sci.* 31 (3), 242–281.
- Kanakasabapathy, P., Swarup, K.S., 2010. Bidding strategy for pumped-storage plant in pool-based electricity market. *Energy Convers. Manag.* 51 (3), 572–579.
- Katsaprakakis, D.A., Kalligeros, S., Pasadakis, N., Moniak, M., Skias, I., 2015. The feasibility of the introduction of natural gas into the electricity production system in the island of Crete (Greece). *Energy Sustain. Develop.* 27, 155–167.
- Khan, I., Jack, M.W., Stephenson, J., 2018. Analysis of greenhouse gas emissions in electricity systems using time-varying carbon intensity. *J. Clean. Prod.* 184, 1091–1101.
- Khan, S.U.-D., Khan, S.U.-D., 2017. Karachi Nuclear Power Plant (KANUPP): as case study for techno-economic assessment of nuclear power coupled with water desalination. *Energy* 127, 372–380.
- Kong, J., Skjelbred, H.L., Fosso, O.B., 2020. An overview on formulations and optimization methods for the unit-based short-term hydro scheduling problem. *Elec. Power Syst. Res.* 178, 106027.
- Koroneos, C., Dompros, A., Roumbas, G., 2007. Renewable energy driven desalination systems modelling. *J. Clean. Prod.* 15 (5), 449–464.
- Larroyd, P.V., de Matos, V.L., Finardi, E.C., 2017. Assessment of risk-averse policies for the long-term hydrothermal scheduling problem. *Energy Syst.* 8 (1), 103–125.
- Lattemann, S., Höpner, T., 2008. Environmental impact and impact assessment of seawater desalination. *Desalination* 220 (1–3), 1–15.
- Li, Z., Lu, L., Lv, P., Du, H., Guo, J., He, X., Ma, J., 2017. Carbon footprints of pre-impoundment clearance on reservoir flooded area in China's large hydro-projects: implications for GHG emissions reduction in the hydropower industry. *J. Clean. Prod.* 168, 1413–1424.
- Liang, H., Liu, Y., Li, F., Shen, Y., 2018. A multiobjective hybrid bat algorithm for combined economic/emission dispatch. *Int. J. Electr. Power Energy Syst.* 101, 103–115.
- Liu, C., Shahidehpour, M., Fu, Y., Li, Z., 2009. Security-constrained unit commitment with natural gas transmission constraints. *IEEE Trans. Power Syst.* 24 (3), 1523–1536.
- Liu, C., Shahidehpour, M., Wang, J., 2010. Application of augmented Lagrangian relaxation to coordinated scheduling of interdependent hydrothermal power and natural gas systems. *IET Gener., Transm. Distrib.* 4 (12), 1314–1325.
- Liu, Y., Tan, S., Jiang, C., 2016. Interval optimal scheduling of hydro-PV-wind hybrid system considering firm generation coordination. *IET Renew. Power Gener.* 11 (1), 63–72.

- López-Salgado, C.J., Helseth, A., Añó, O., Ojeda-Esteybar, D.M., 2020. Stochastic daily hydrothermal scheduling based on decomposition and parallelization. *Int. J. Electr. Power Energy Syst.* 118, 105700.
- Lu, N., Chow, J.H., Desrochers, A.A., 2003. Pumped-storage hydro-turbine bidding strategies in a competitive electricity market. In: *Power Engineering Society General Meeting, 2003, IEEE*. IEEE, pp. 831–837.
- Majidi, M., Nojavan, S., Zare, K., 2017. A cost-emission framework for hub energy system under demand response program. *Energy* 134, 157–166.
- Majidi, M., Nojavan, S., Zare, K., 2019a. Optimization Framework Based on Information Gap Decision Theory for Optimal Operation of Multi-Energy Systems, *Robust Optimal Planning and Operation of Electrical Energy Systems*. Springer, pp. 35–59.
- Majidi, M., Nojavan, S., Zare, K., 2019b. Risk-based performance of multi-carrier energy systems: robust optimization framework. In: *Robust Optimal Planning and Operation of Electrical Energy Systems*. Springer, pp. 271–291.
- Manju, S., Sagar, N., 2017. Renewable energy integrated desalination: a sustainable solution to overcome future fresh-water scarcity in India. *Renew. Sustain. Energy Rev.* 73, 594–609.
- Mao, G., Wang, S., Teng, Q., Zuo, J., Tan, X., Wang, H., Liu, Z., 2017. The sustainable future of hydropower: a critical analysis of cooling units via the theory of inventive problem solving and life cycle assessment methods. *J. Clean. Prod.* 142, 2446–2453.
- Mariano, J.D.A., Santos, F.R., Brito, G.W., Junior, J.U., Junior, E.F.C., 2016. Hydro, thermal and photovoltaic power plants: a comparison between electric power generation, environmental impacts and CO₂ emissions in the Brazilian scenario. *Int. J. Energy Environ.* 7 (4), 347.
- Martinez-Mares, A., Fuerte-Esquivel, C.R., 2012. A unified gas and power flow analysis in natural gas and electricity coupled networks. *IEEE Trans. Power Syst.* 27 (4), 2156–2166.
- Matejcek, L., 2017. *Hydropower: Assessment of Energy Potential and Environmental Issues in the Local and Global Scales, Assessment of Energy Sources Using GIS*. Springer, pp. 165–193.
- Mehrjerdi, H., 2019. Optimal correlation of non-renewable and renewable generating systems for producing hydrogen and methane by power to gas process. *Int. J. Hydrogen Energy* 44 (18), 9210–9219.
- Mesfun, S., Sanchez, D.L., Leduc, S., Wetterlund, E., Lundgren, J., Biberacher, M., Kraxner, F., 2017. Power-to-gas and power-to-liquid for managing renewable electricity intermittency in the Alpine Region. *Renew. Energy* 107, 361–372.
- Minutillo, M., Perna, A., 2014. Renewable energy storage system via coal hydro-gasification with co-production of electricity and synthetic natural gas. *Int. J. Hydrogen Energy* 39 (11), 5793–5803.
- Mirzaei, M.A., Nazari-Heris, M., Mohammadi-Ivatloo, B., Zare, K., Marzband, M., Anvari-Moghaddam, A., 2020. A novel hybrid framework for Co-optimization of power and natural gas networks integrated with emerging technologies. *IEEE Syst. J.* <https://doi.org/10.1109/JSYST.2020.2975090>.
- Moh'd A, A.-N., Al-Ammari, W.A., 2020. A novel hybrid and interactive solar system consists of Stirling engine vacuum evaporator thermoelectric cooler for electricity generation and water distillation. *Renew. Energy* 153, 1053–1066.
- Mohibullah, M., Radzi, A.M., Hakim, M.I.A., 2004. In: *Basic Design Aspects of Micro Hydro Power Plant and its Potential Development in Malaysia, Power and Energy Conference, 2004, PECON 2004, Proceedings. National. IEEE*, pp. 220–223.
- Murugan, A., de Huu, M., Bacquart, T., van Wijk, J., Arrhenius, K., te Ronde, I., Hemfrey, D., 2019. Measurement challenges for hydrogen vehicles. *Int. J. Hydrogen Energy* 44 (35), 19326–19333.
- Nadakuditi, G., Sharma, V., Naresh, R., 2016. Application of non-dominated sorting gravitational search algorithm with disruption operator for stochastic multi-objective short term hydrothermal scheduling. *IET Gener., Transm. Distrib.* 10 (4), 862–872.
- Nakhmkin, M., 2012. Advanced adiabatic compressed air energy storage system. *Google Patents*, WO201159586A3.
- Nayar, K.G., Lienhard, V.J.H., 2020. Brackish water desalination for greenhouse agriculture: comparing the costs of RO, CCRO, EDR, and monovalent-selective EDR. *Desalination* 475, 114188.
- Nazari-Heris, M., Babaei, A.F., Mohammadi-Ivatloo, B., Asadi, S., 2018a. Improved harmony search algorithm for the solution of non-linear non-convex short-term hydrothermal scheduling. *Energy* 151, 226–237.
- Nazari-Heris, M., Mirzaei, M.A., Mohammadi-Ivatloo, B., Marzband, M., Asadi, S., 2020. Economic-environmental effect of power to gas technology in coupled electricity and gas systems with price-responsive shiftable loads. *J. Clean. Prod.* 244, 118769.
- Nazari-Heris, M., Mohammadi-Ivatloo, B., 2017. Design of Small Hydro Generation Systems. *Distributed Generation Systems: Design, Operation and Grid Integration*, p. 301.
- Nazari-Heris, M., Mohammadi-Ivatloo, B., Asadi, S., 2018b. Robust stochastic optimal short-term generation scheduling of hydrothermal systems in deregulated environment. *J. Energy Syst.* 2 (4), 168–179.
- Nazari-Heris, M., Mohammadi-Ivatloo, B., Gharehpetian, G., 2017a. Short-term scheduling of hydro-based power plants considering application of heuristic algorithms: a comprehensive review. *Renew. Sustain. Energy Rev.* 74, 116–129.
- Nazari-Heris, M., Mohammadi-Ivatloo, B., Haghighi, A., 2017b. Optimal short-term generation scheduling of hydrothermal systems by implementation of real-coded genetic algorithm based on improved Mühlhenn mutation. *Energy* 128, 77–85.
- Novosel, T., Čosić, B., Pukšec, T., Krajačić, G., Duić, N., Mathiesen, B.V., Lund, H., Mustafa, M., 2015. Integration of renewables and reverse osmosis desalination—Case study for the Jordanian energy system with a high share of wind and photovoltaics. *Energy* 92, 270–278.
- Nuez, I., Garcia Latorre, F.J., 2016. Reducing costs of electric power generation through the integration of desalinated water production into insular weak electric systems. *Desalin. Water Treat.* 57 (4), 1623–1633.
- Nunes, J.B., Mahmoudi, N., Saha, T.K., Chattopadhyay, D., 2018. A stochastic integrated planning of electricity and natural gas networks for Queensland, Australia considering high renewable penetration. *Energy* 153, 539–553.
- Odetayo, B., MacCormack, J., Rosehart, W.D., Zareipour, H., 2017. A sequential planning approach for Distributed Generation and natural gas networks. *Energy* 127, 428–437.
- Oliveira, T.V., Nogueira, A.A., 2016. Analysis of development of carbonation and surface wear of the concrete: a case study in Ship Lock 1 of the transposition system of Tucuruí dam. *J. Build. Pathol. Rehabil.* 1 (1), 17.
- Padmini, S., Jegatheesan, R., 2016. A new model for short-term hydrothermal scheduling of a GENCO in the competitive electricity market. *Ind. J. Sci. Technol.* 9 (46).
- Paish, O., 2002. Small hydro power: technology and current status. *Renew. Sustain. Energy Rev.* 6 (6), 537–556.
- Penche, C., 2004. *Guide on How to Develop a Small Hydropower Plant*. European Small Hydropower Association.
- Pérez-Díaz, J.I., Jiménez, J., 2016. Contribution of a pumped-storage hydropower plant to reduce the scheduling costs of an isolated power system with high wind power penetration. *Energy* 109, 92–104.
- Pradhan, M., Roy, P.K., Pal, T., 2020. Multi-objective short-term hydro-thermal scheduling using meta-heuristic approaches. In: *Handbook of Research on Advancements of Swarm Intelligence Algorithms for Solving Real-World Problems*. IGI Global, pp. 382–414.
- Purohit, P., 2008. Small hydro power projects under clean development mechanism in India: a preliminary assessment. *Energy Pol.* 36 (6), 2000–2015.
- Qadrdan, M., Abeysekera, M., Chaudry, M., Wu, J., Jenkins, N., 2015. Role of power-to-gas in an integrated gas and electricity system in Great Britain. *Int. J. Hydrogen Energy* 40 (17), 5763–5775.
- Qadrdan, M., Ameli, H., Strbac, G., Jenkins, N., 2017. Efficacy of options to address balancing challenges: integrated gas and electricity perspectives. *Appl. Energy* 190, 181–190.
- Qiao, Z., Guo, Q., Sun, H., Pan, Z., Liu, Y., Xiong, W., 2017. An interval gas flow analysis in natural gas and electricity coupled networks considering the uncertainty of wind power. *Appl. Energy* 201, 343–353.
- Qiu, J., Dong, Z.Y., Zhao, J.H., Meng, K., Zheng, Y., Hill, D.J., 2015a. Low carbon oriented expansion planning of integrated gas and power systems. *IEEE Trans. Power Syst.* 30 (2), 1035–1046.
- Qiu, J., Dong, Z.Y., Zhao, J.H., Xu, Y., Zheng, Y., Li, C., Wong, K.P., 2015b. Multi-stage flexible expansion co-planning under uncertainties in a combined electricity and gas market. *IEEE Trans. Power Syst.* 30 (4), 2119–2129.
- Qiu, J., Zhao, J., Yang, H., Wang, D., Dong, Z.Y., 2018. Planning of solar photovoltaics, battery energy storage system and gas micro turbine for coupled micro energy grids. *Appl. Energy* 219, 361–369.
- Quteishat, K., 2009. *Energy for Desalination*. Proceedings of World Water Week in Stockholm, Stockholm, Sweden, pp. 16–22.
- Rasku, T., Miettinen, J., Rinne, E., Kiviluoma, J., 2020. Impact of 15-day energy forecasts on the hydro-thermal scheduling of a future Nordic power system. *Energy* 192, 116668.
- Roberts, B., 2009. Capturing grid power. *IEEE Power Energy Mag.* 7 (4).
- Roy, P.K., Sur, A., Pradhan, D.K., 2013. Optimal short-term hydro-thermal scheduling using quasi-oppositional teaching learning based optimization. *Eng. Appl. Artif. Intell.* 26 (10), 2516–2524.
- Rozali, N.E.M., Alwi, S.R.W., Manan, Z.A., Klemes, J.J., Hassan, M.Y., 2014. Optimal sizing of hybrid power systems using power pinch analysis. *J. Clean. Prod.* 71, 158–167.
- Sahin, C., Li, Z., Shahidepour, M., Erkmen, I., 2011. Impact of natural gas system on risk-constrained midterm hydrothermal scheduling. *IEEE Trans. Power Syst.* 26 (2), 520–531.
- Sahoo, U., Kumar, R., Pant, P., Chaudhury, R., 2015. Scope and sustainability of hybrid solar–biomass power plant with cooling, desalination in polygeneration process in India. *Renew. Sustain. Energy Rev.* 51, 304–316.
- Salehi-Dobakhshari, A., Fotuhi-Firuzabad, M., 2011. Integration of large-scale wind farm projects including system reliability analysis. *IET Renew. Power Gener.* 5 (1), 89–98.
- Sebri, M., 2015. Use renewables to be cleaner: meta-analysis of the renewable energy consumption—Economic growth nexus. *Renew. Sustain. Energy Rev.* 42, 657–665.
- Shahzad, M.W., Burhan, M., Ang, L., Ng, K.C., 2017. Energy–water–environment nexus underpinning future desalination sustainability. *Desalination* 413, 52–64.
- Shannon, M.A., Bohn, P.W., Elimelech, M., Georgiadis, J.G., Mariñas, B.J., Mayes, A.M., 2008. Science and technology for water purification in the coming decades. *Nature* 452 (7185), 301–310.
- Sheikhi, A., Bahrami, S., Ranjbar, A.M., 2015. An autonomous demand response program for electricity and natural gas networks in smart energy hubs. *Energy* 89, 490–499.
- Skjelbred, H.I., Kong, J., Fosso, O.B., 2020. Dynamic incorporation of nonlinearity into MILP formulation for short-term hydro scheduling. *Int. J. Electr. Power Energy Syst.* 116, 105530.
- Soroudi, A., 2013. Robust optimization based self scheduling of hydro-thermal Genco in smart grids. *Energy* 61, 262–271.

- Srithar, K., Rajaseenivasan, T., Karthik, N., Periyannan, M., Gowtham, M., 2016. Stand alone triple basin solar desalination system with cover cooling and parabolic dish concentrator. *Renew. Energy* 90, 157–165.
- Tarasak, P., Chai, C.C., Kwok, Y.S., Oh, S.W., 2014. Demand bidding program and its application in hotel energy management. *IEEE Trans. Smart Grid* 5 (2), 821–828.
- Tassinari, C., Bonilla, S., Agostinho, F., Almeida, C., Giannetti, B., 2016. Evaluation of two hydropower plants in Brazil: using emergy for exploring regional possibilities. *J. Clean. Prod.* 122, 78–86.
- Thiel, G.P., Miller, J.A., Zubair, S.M., Lienhard, J.H., 2013. Effect of mass extractions and injections on the performance of a fixed-size humidification–dehumidification desalination system. *Desalination* 314, 50–58.
- Torcellini, P.A., Long, N., Judkoff, R., 2003. Consumptive Water Use for US Power Production. National Renewable Energy Laboratory Golden, CO.
- Tsujimura, T., Suzuki, Y., 2019. Development of a large-sized direct injection hydrogen engine for a stationary power generator. *Int. J. Hydrogen Energy* 44 (22), 11355–11369.
- Unsihuay-Vila, C., Marangon-Lima, J., de Souza, A.Z., Perez-Arriaga, I.J., Balestrassi, P.P., 2010. A model to long-term, multiarea, multistage, and integrated expansion planning of electricity and natural gas systems. *IEEE Trans. Power Syst.* 25 (2), 1154–1168.
- Üster, H., Dilaveroglu, S., 2014. Optimization for design and operation of natural gas transmission networks. *Appl. Energy* 133, 56–69.
- van Leeuwen, C., Mulder, M., 2018. Power-to-gas in electricity markets dominated by renewables. *Appl. Energy* 232, 258–272.
- Venetsanos, K., Angelopoulou, P., Tsoutsos, T., 2002. Renewable energy sources project appraisal under uncertainty: the case of wind energy exploitation within a changing energy market environment. *Energy Pol.* 30 (4), 293–307.
- Wa'el A, H., Memon, F.A., Savic, D.A., 2018. A risk-based assessment of the household water-energy-food nexus under the impact of seasonal variability. *J. Clean. Prod.* 171, 1275–1289.
- Wang, H., Zhang, H., Gu, C., Li, F., 2017. Optimal design and operation of CHPs and energy hub with multi objectives for a local energy system. *Energy Procedia* 142, 1615–1621.
- Warland, G., Henden, A.L., Mo, B., 2016. Use of parallel processing in applications for hydro power scheduling—current status and future challenges. *Energy Procedia* 87, 157–164.
- Wu, L., Shahidehpour, M., Li, Z., 2008. GENCO's risk-constrained hydrothermal scheduling. *IEEE Trans. Power Syst.* 23 (4), 1847–1858.
- Yang, Z., Liu, P., Cheng, L., Wang, H., Ming, B., Gong, W., 2018. Deriving operating rules for a large-scale hydro-photovoltaic power system using implicit stochastic optimization. *J. Clean. Prod.* 195, 562–572.
- Yuan, L., Zhou, J., Li, C., Xie, M., Mo, L., 2016. Benefit and risk balance optimization for stochastic hydropower scheduling. *Water Resour. Manag.* 30 (10), 3347–3361.
- Yuan, L., Zhou, J., Mai, Z., Li, Y., 2017. Random fuzzy optimization model for short-term hydropower scheduling considering uncertainty of power load. *Water Resour. Manag.* 31 (9), 2713–2728.
- Yuan, Z., He, S., Alizadeh, A.a., Nojavan, S., Jermisittiparsert, K., 2020. Probabilistic scheduling of power-to-gas storage system in renewable energy hub integrated with demand response program. *J. Energy Storage* 29, 101393.
- Zeng, Q., Zhang, B., Fang, J., Chen, Z., 2017. A bi-level programming for multistage co-expansion planning of the integrated gas and electricity system. *Appl. Energy* 200, 192–203.
- Zhang, H., Zhou, J., Zhang, Y., Fang, N., Zhang, R., 2013. Short term hydrothermal scheduling using multi-objective differential evolution with three chaotic sequences. *Int. J. Electr. Power Energy Syst.* 47, 85–99.
- Zhang, X., Che, L., Shahidehpour, M., Alabdulwahab, A., Abusorrah, A., 2016a. Electricity-natural gas operation planning with hourly demand response for deployment of flexible ramp. *IEEE Trans. Sustain. Energy* 7 (3), 996–1004.
- Zhang, X., Che, L., Shahidehpour, M., Alabdulwahab, A.S., Abusorrah, A., 2017. Reliability-based optimal planning of electricity and natural gas interconnections for multiple energy hubs. *IEEE Trans. Smart Grid* 8 (4), 1658–1667.
- Zhang, X., Shahidehpour, M., Alabdulwahab, A., Abusorrah, A., 2016b. Hourly electricity demand response in the stochastic day-ahead scheduling of coordinated electricity and natural gas networks. *IEEE Trans. Power Syst.* 31 (1), 592–601.
- Zhang, X., Shahidehpour, M., Alabdulwahab, A.S., Abusorrah, A., 2015b. Security-constrained co-optimization planning of electricity and natural gas transportation infrastructures. *IEEE Trans. Power Syst.* 30 (6), 2984–2993.
- Zhang, X., Vesselinov, V.V., 2016. Energy-water nexus: balancing the tradeoffs between two-level decision makers. *Appl. Energy* 183, 77–87.
- Zhou, H., Zheng, J.H., Li, Z., Wu, Q.H., Zhou, X.X., 2019. Multi-stage contingency-constrained co-planning for electricity-gas systems interconnected with gas-fired units and power-to-gas plants using iterative Benders decomposition. *Energy* 180, 689–701.
- Zou, Q., Liu, X., 2016. Economic effects analysis of seawater desalination in China with input–output technology. *Desalination* 380, 18–28.